



STATEMENT OF TECHNICAL REQUIREMENTS

FOR

**FOR FIRE SURVIVAL, HIGH TEMPERATURE ZONE AND FIRE
RETARDANT HALOGEN FREE SHEATHED ELECTRON BEAM
CROSSLINKED ELECTRIC CABLES**

MAR 12

EE-50-13 (REV 1)

**MATERIAL BRANCH
DIRECTORATE OF ELECTRICAL ENGINEERING
INTEGRATED HEADQUARTERS
MOD (NAVY)
'D' WING, SENA BHAWAN
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(i) RECORD OF AMENDMENTS

SL NO	AMENDMENT	AUTHORITY	DATE	SIGNATURE
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II. REVISION NOTE

No revision incorporated.

III. HISTORICAL RECORD

This standard supersedes the following:-

NIL.

CONDITIONS OF RELEASE

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8. Unless otherwise specified, reference in this specification to any document means the issue and all amendments to the document current at the date of issue or subsequent amendments to this specification.

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INTRODUCTION

1. This EED-50-13 Electrical Specification is applicable to all electrical cables with extruded and taped insulated flame retardant halogen free Sheathed (i.e. minimum flame spread, low smoke emission and limited toxic fume emission) Electron Beam Cross linked cables with fire survival characteristics. The specification is relevant to cables for use onboard Indian Naval Ships and Crafts and is sponsored by the Indian Navy, Directorate of Electrical Engineering, Naval Headquarters, Sena Bhawan, New Delhi - 110 011
2. The manufacturers of Cables for applications on Naval ships should be **ISO 9001 –2000/ ISO 14001** certified organizations.
3. It is to be applied as required for the manufacture, supply and testing of electric cables to be used on Indian Naval ships
4. If it is found to be technically unsuitable for any particular requirement, the Sponsor is to be informed in writing of the circumstances with a copy to Directorate of Electrical Engineering, Naval Headquarters, Sena Bhawan, New Delhi-110 011
5. Any user of this Specification may propose an amendment to it. Proposals for amendments which are:-
 - (a) Not directly applicable to a particular contract are to be made to the Sponsor of the EED-50-13.
 - (b) Directly applicable to a particular contract are to be dealt with using existing procedures or as specified in the contract.
6. No alteration is to be made to this EED-Q specification except by the issue of a formal amendment by Director of Electrical Engineering, Naval Headquarters.
7. Unless otherwise stated, reference in this EED-Q specification, to “approval”, “approved”, “authorized” or similar terms means “by the Naval Headquarters”.
8. When this EED-Q is specified in the contracts /purchase orders users are responsible for their correct application and for complying with contract and any statutory requirements. Compliance with this EED-Q does not of itself confer immunity from legal obligations.

SECTION-I

SCOPE

0101. This Specification covers the design, manufacture and testing requirements of Electron Beam Cross linked, extruded and taped insulated fire retardant halogen free sheathed cables with fire survival characteristics for power, lighting, control and communication and instrumentation circuits in Surface Ships and submarines.

0102. The cables are rated for 250, 440 and 600 V (rms sinusoidal) between cores and earth / screens and be suitable for use on systems in which the frequency of supply is up to and including 400 Hz.

0103. Cables will be used in applications where fuel and lubrication oils, hydraulic fluids and water are present, and are to be suitable for use within the temperature range -30°C to 120°C . The maximum continuous conductor temperature of cable can be up to 120°C . The lower temperature limit refers to weather deck applications under 'arctic' conditions, and the maximum continuous conductor temperature to the combination of ambient temperature and temperature rise due to maximum current flow. The insulation and sheathing materials are expected to have a service life exceeding 40000 hours at the maximum continuous conductor temperature before deterioration as a direct result of thermal ageing, necessitating cable replacement.

0104. Cables are required to withstand the normal installation hazards like, conditions of riveting, temporary support and glanding into equipment, banding and clipping that are accepted practices in the shipbuilding and ship refitting industry.

0105. These cables may be used in flexing and fixed application subject to the following minimum bending radius -

- (a) Flexing application: - 10 times the overall diameter of the cable.
- (b) Fixed application: - 4 times the overall diameter of cable.

SECTION II

RELATED DOCUMENTS

0201. The following documentation or the latest in effect is to form a part of this specification to the extent specified herein, except where a specific issue is indicated. In case of a conflict between the contents of this document and the applicable portions of the referenced documents, the contents of this document shall take precedence.

0202. Reference to following documents is made in this EED:-

<u>SL.NO.</u>	<u>DOCUMENT</u>	<u>DESCRIPTION</u>
a.	ASTM D 2863	: Standard Test method for measuring the minimum Oxygen concentration to support Candle like combustion of Plastics (Oxygen Index)
b.	ASTM D-2843	: Standard test method for specific optical density of smoke generated by solid materials
c.	ASTM D 3677	: Standard test method for rubber – identification by infrared spectrophotometer
d.	BS 6469	: Methods of test for insulation and sheath of electric cables.
e.	BS 5099	: Spark testing of Electric Cables
f.	BS 6899	: Rubber insulation and sheath of electric Cables
g.	BS 419	: Specification for varnished fabrics and tapes for electrical purposes
h.	BS 6360	: Conductors in insulated cables and cords.
i.	BS 6387	: Requirements to maintain circuit integrity under fire conditions
j.	BS 5691 part 1	: Determination of thermal endurance properties of electrical insulating materials
k.	BS G 230	: Specification for general requirements for aircraft electrical cables
l.	BS 1142 part 2	:
m.	BS 2011	: Basic Environmental testing procedures
n.	IEC –228	: Conductors of Insulated cables
o.	IEC –811-1-1	: Common test methods for insulating and sheathing materials of electric cables- Measurement of thickness and overall dimensions.

<u>SL.NO.</u>	<u>DOCUMENT</u>	<u>DESCRIPTION</u>
p.	IEC –811-2-1	: Common test methods for insulating and sheathing materials of electric cables – Ozone resistance test- Hot set test – Mineral oil immersion test.
q.	IEC –811-3-1	: Common test methods for insulating and sheathing materials of electric cables – Pressure test at high temperature – Test for resistance to cracking
r.	IEC –811-1-3	: Common test methods for insulating and sheathing materials of electric cables- Methods for determining the density – Water absorption tests – Shrinkage test.
s.	IEC – 216 –1	: Guide for determination at thermal endurance properties of electrical insulating materials. Part– 1: General Guideline for ageing procedures and evaluation of test results
t.	IEC – 332-1	: Test on electric cables under fire conditions: Part – 1: Test on a single vertical insulated wire or cable
u.	IEC – 754 – 2	: Determination of degree of acidity of gasses evolved during the combustion of materials taken from electric cables by measuring pH and conductivity
v.	IEC-1034-2	: Measurement of Smoke Density of cables burning under defined conditions; Part-2 “Test procedure and requirements”.
w.	IS 10810	: Methods Of Test For Cables
x.	IS 8130	: Conductors for Insulated Electric Cables and Flexible Cords
y.	IEC 68-2-20	: Method 1 Basic environmental testing procedures
z.	IS 10418	: Specifications for packing drums.
aa.	IEEE 383	: Type Test of Class I Electric Cables, Field Splices and Connections For Nuclear Power Generating Stations.
aa	MIL-DTL-27500H	: Cable, Power Electrical and special purpose electrical shielded and unshielded, General specification
ab	MIL-C-13777	: Cables Special purpose, Electrical.
ac	MIL-DTL-24643	: Cables And Cords, Low-Smoke, Electric, For Shipboard Use General Specification
ad	NES 713	: Determination of the Toxicity Index of the products of combustion from small specimens of Materials.

<u>SL.NO.</u>	<u>DOCUMENT</u>	<u>DESCRIPTION</u>
ae	Def Stan 02- : 527 (NES 527)	Requirements for cables Electric, fire survival, High Temperature zones and Limited Fire Hazard sheathed
af	DEF STAN 61- : 12 part 31	wires, cords, & cables, Electrical- metric units, part 31: sheath – limited fire hazard
ag	NEMA WC 3 :	Rubber insulated wire and cable for transmission and distribution of electrical energy.
ah	NEMA WC 5 :	Thermoplastic insulated wire and cable for transmission and distribution of electrical energy
ai	VG 95218 Part : 2	Electrical cables and insulated wires

SECTION III

REQUIREMENTS FOR MATERIALS

Conductor

0301. The conductor shall be flexible (class 5) and made up with circular Electroplated annealed tinned copper wires unless otherwise stated in the Tables of Dimensions and is to comply in all respects with BS 6360 or IS: 8130- 1984 or IEC-228, except that the conductor for cables covered under tables C9, C10 and C11 shall be class 2. The wires taken from completed cable are not required to pass the tinning test. Nominal cross – sectional area, nominal diameter of wires in conductor, number of wires and maximum resistance of conductor at 20⁰ C shall be as specified in respective tables.

0302 For non flexible conductors, the diameter of individual wires shall be nominal and the number of individual wires is as stated.

0303 For flexible conductors, the diameter and the number of individual wires shall be nominal.

0304 The conductor shall be concentric stranded below 10-Sqmm and rope stranded 10 mm² and above.

Separator Tape

0305. Separator tape shall be polyethylene terephthalate (PTP) or other non-hygroscopic tape & may be applied over the conductor. Nominal thickness of tape is 0.020 mm to 0.050 mm.

Conductor Insulation

0306. Conductor insulation is to comply with the following:-

(a) For cores with nominal cross section of conductor up to and including 16 mm² Cross linked Silicone rubber compound to comply with the following parameters:-

(i) Minimum Tensile Strength (in accordance with Clause 0505) of 8.0 N/mm².

(ii) Minimum Elongation at Break (in accordance with Clause 0505) of 150%.

(iii) Residual Elongation at break after ageing at 150°C for 168 hrs (in accordance with Clause 0505) of 75% retained.

(b) For cores with nominal cross section of conductor greater than 16 mm² as listed in Table C1 to C3 inc. in Annex C Cross linked Silicone rubber compound to comply

with above parameters.

(c) For cables requiring to meet the fire survival test requirements, as per para 0542 and 0543, the insulation will additionally consist of glass mica tape, irrespective of the nominal cross section of the conductor. The minimum thickness of the mica glass tape shall be 0.075 mm.

0307 The manufacturer shall establish the Infra Red Spectrograph for rubber and furnish the same to DQA (N). In case of any change in material, the manufacturer shall obtain the approval of IHQ MOD/DEE before adoption. The manufacturer shall establish test facilities for infra red (IR) spectrography at their works.

0308. Braid / Protective tape(s): The glass braid is to be a continuous filament non alkaline yarn applied closely woven. The protective tape(s) should be treated glass tape with a minimum thickness of 0.050 mm. The minimum thickness of the mica glass tape shall be 0.075 mm.

0309. Lacquer: The lacquer is to be halogen free and is not to have any detrimental effect on other materials used in the construction of the cable.

Wrapping Tapes

0310. Wrapping Tapes, other than conductor separator tapes, are to be applied helically with a minimum of 10% overlap and any tape used shall be removable without damage to the insulation in the completed cable. The width of the tape and the angle of application shall be such that it minimizes wrinkles and creases when the cable is bent. The tape should be PTP film tape(s) with minimum thickness 0.020 mm (or) closely woven cotton tape without selvedge and proofed on one side with an elastomeric compound;

Filler(s), Binder(s) and Innersheath

0311. Where filler(s), binder(s) or inner sheath are considered necessary, these are to be non-hygroscopic halogen free, low toxicity and flame retardant. The filler(s), binder(s) or Inner sheath shall have no deleterious effect on the components or flexibility of the cable. The flame retardant capability of the filler will be evaluated during flammability testing.

Electrical Screen

0312. Where specified the screen should be a symmetrical braiding of Electroplated annealed tinned copper wire of diameter given in the Tables of Dimensions.

Sheath

0313. The sheath used in the construction of cables covered in this Document is to be fire retardant low smoke zero halogen Electron Beam cross linked special polyolefin EMA/EVA/EEA type compound in accordance with test requirement mentioned in the specification. The material of sheath shall conform to various requirements and tests in accordance with in general to NES 518 and shall have life exceeding 40000 hours at rated current capacity at 120⁰C. It is to be black in colour and compatible with all other materials employed in the cable construction and also other sheathing materials as tested. The material should comply to the following parameters:-

TEST DETAILS	PARAMETER
Critical Oxygen Index	29 min (ASTM-D-2863)
Temperature Index	250 ⁰ C min (ASTM-D-2863)
Halogen Content	Negative result for Halogens (IEC-754-1)
Toxicity Index	5 max (NES-713)
Tensile Strength	>8 Mpa min
Elongation at Break	250%
Hot Set Test	100% Max Elongation 25% Max Permanent Elongation
Resistance to Fluids	As per Annex-D
Tear Resistance	5 N/mm min
Thermal endurance	40000 hours at 120 ⁰ C (IEC- 216-1)
Accelerated Ageing	As per specified in Annex-D
Smoke Density Rating	20 max (ASTM-D-2843)
Resistance to Fluids	As per Annex-D
Ozone Resistance	No cracks
Cold Elongation	20% min elongation
Heat Shock	No cracks
Insulation Resistance Constant	0.1 M Ω Km.
Ageing in air oven at 150 ⁰ C +/-3 ⁰ C for 10 days.	+/- 30% Max. Variation in tensile strength and elongation at break.
Acid Gas generation/ corrosivity of combustion gases.	pH Index $\geq$ 4.3 , Electrolytic Conductivity $\leq$ 100 micro.S/ cm

Table 3.1 Acceptance Test Limits

0314. Where materials other than those above are used to meet the required performance criteria then, apart from undergoing the qualification approval tests as specified in Table 3.1. due consideration must be given to their contribution to levels of smoke, toxicity and acid gas generated by the cable when subjected to excessive temperatures or fire.

SECTION IV

CONSTRUCTION

0401. **Electron Beam Cross Linking.** Cross linking is the linking of macro molecules into a network by main valence connection, wherein the physical and chemical properties of material are changed. Electron beam cross linking is a process employed for curing of insulation and sheathing for cables wherein an electron beam radiation is used for cross linking.

0402. Rubber sheath being weak, soluble and fusible are required to be cross-linked to convert it into strong, insoluble and infusible thermoset. Electron Beam Cross-linking method should be used for cross-linking to achieve properties mentioned herein. Rubbers should be adequately formulated with efficient antioxidant, fire retardant & smoke suppressant for the cable to be suitable for 120°C continuous operation, low smoke, halogen free and toxic gas emission. The electron beam Cross linking should transform the rubber (thermoplastic) molecules of both insulation and sheath compound into three dimensional network structure which should restrains the movement of macro-molecular chain molecules of strong chemical bond completely insoluble and infusible compound (thermoset). The cross linked cables should have following distinct design/operational features:-

- (a) Higher temperatures withstand capability up to 120⁰ C.
- (b) Halogen free & non-toxic emission.
- (c) Reduced dimensions by using comparatively thinner insulation/sheath to achieve same properties with increased current ratings.
- (d) Better flexibility and hence have smaller bending radius.

Conductor Stranding

0403. The number and nominal diameter of the individual wires to be used to form a conductor are to be as stated in the Tables of Dimensions. The conductor shall be concentric stranded below 10-mm² and rope stranded 10 mm² and above with uni-directional lay. In both the cases, the unidirectional lay shall be left hand. Further, the conductor resistance values should be met. The resistance of each conductor is to be measured at room temperature and corrected to 20 Deg C by the constants specified in IEC-60228 / IS 8130. The corrected resistance is not to exceed the values stated in the tables of dimensions. The conductor wires are to be free from discolouration, dulling and bluing.

0404 For non flexible conductors, the diameter of individual wires shall be nominal and the number of individual wires is as stated.

0405 For flexible conductors, the diameter and the number of individual wires shall be nominal.

Separator tape

0406. A separator tape may be applied over the conductor. Embrittlement or discolouration of the tape after cross-linking of the core may be ignored.

Conductor Insulation

0407. Insulation used in the construction of the cables is to meet the requirements of Annexure D. The average radial thickness of the insulation system is to be not less than the values stated in the Tables of Dimensions and the thickness at any point is not to fall below the tabulated value by more than 20%.

0408. A protective layer of closely woven glass braid or tape is to be applied over the extruded insulation. The angle of lay of the glass braid is to be such that all other requirements of this Document can be met. The angle between the glass thread and the conductor axis is to be as closely as possible to 55° in all cases where applicable except for miniature cables listed in Tables C 9, C 10 and C 11. In case of protective tape(s) the minimum overlap should be 10%.

0409. The protective layer is to be coated with a transparent lacquer so that reasonable handling will not fray or loosen fibers at cut ends. The lacquer is not to obscure the core identification except in those cases where coloured lacquer is used.

0410 If the insulated cores are taped then, the insulation is to consist of not less than 3 tapes, and may be suitably lubricated with slippery compound at the manufacturer's discretion. The tape is to be applied helically, in the same direction, successive layers breaking joints so as to form a reasonably smooth and compact covering of uniform thickness.

0411. To meet the optional fire test requirements, the Insulation system used in the construction of the cables to this specification may use mica glass tape. The thickness at any point of the insulation system is not to fall below the tabulated value by more than 20%.

Identification of cores

0412 The colour of insulation shall be natural/ black. Individual cores shall be identified by means of contrast colour numbers printed on the outer layer of core insulation. The figures are to be well defined and permanent , and applied at intervals not greater than 150 mm. Measures are to be taken to render the numerals 3 and 8 also 6 and 9 readily distinctive by the shapes of the figures, base lines or suffix letters. If the last method be adopted the identifying initials are to be omitted where numbers contain more than one figure. Whilst the identification of cores by numbering is preferred, colour coding of cores may also be accepted if found necessary for manufacturing reasons. In both the coding systems the identification is to remain legible.

0413 For cables with greater number of cores, pairs, triads or quads the numbering is to be similarly extended, starting from the centre core, pair, triad or quad or first layer and proceeding in advancing order in the same direction in each layer and in sequence from the preceding layer or centre. The numbering/ colour coding scheme should be as follows:-

SL. No.	NO. OF CORES	NUMBERING OF CORES	COLOUR CODING
(a)	Two Cores	One of the core outer insulation should be printed as 1 and the other as 2 or by way of number printed tape	One of the core insulation to be coloured as Red and the other as Blue
(b)	Three cores	Number 1,2 & 3 be printed on outer insulation of respective core or by way of number printed tape	Three cores are to be coloured Red, Blue, and White.
(c)	multi core cables	Number 1, 2, 3, 4 etc. be printed on outer insulation of respective core or by way of number printed tape	Two adjacent cores (counting and direction core) in each layer Red & Blue, remaining cores White.
(d)	Multi pair cables	Cores numbered 1&2 for 1 st pair and 3 & 4 for 2 nd pair etc. or by way of pair number printed tape over pair.	1 pair: Red/Blue Others: Red /Black and Blue /Black marker pairs adjacent in each layer with remaining pairs white / Black. Note: When one pair is used as a centre, the colours are to be white /Black.

(e)	Multi Triad cables	Cores numbered 1, 2, 3 & for 1 st Triad and 4, 5 & 6 for 2 nd triads or by way of triad number printed tape over triad.	Each Triad to be Red/Blue/White and lapped with a suitable numbered tape, or the white core in each triad numbered.
(f)	Multi quad cables	Cores numbered 1, 2, 3, & 4 for 1 st quad and 5, 6, 7 & 8 for 2 nd quad or by way of quad number printed tape over quad.	Each quad to be Red/Blue/White/Black and lapped with a suitable numbered tape, or the white core in each quad numbered.

Laying Up of Cores

0414 For multi core cables the length of lay of the cores in each layer is to be not more than 24 times the diameter over their layer. The direction of the first layer of cores is to be left hand and subsequent layers are to be in alternate directions.

0415 For multi pair and multi triad cables:-

- (a) The insulated cores are to be twisted together with a lay not greater than 160 mm, but it is not considered necessary to have different twist ratios for each pair or triad. The lay lengths of the pairs are to be chosen and related to reduce inductive effect and cross talk to a minimum.
- (b) Cables with a centre consisting of one pair or triad are to have the first layer of pairs or triads laid up with a left hand lay. Subsequent layers are to be in alternate directions.
- (c) Cables with a centre consisting of more than one pair or triad are to be laid up left hand and the first layer applied with a right hand lay. Any Subsequent layers are to be in alternate directions.

0416. The interstices between cores are to be filled with suitable elastomer where necessary which may be cellular. A cellular filler is preferred but the selection of type of filler necessary to meet the requirements of this Document to be left to the manufacturer's discretion. The filler is to be capable of removal from the cores after which the legibility of the identification markings is to be unimpaired. Some adhesion of the filler is acceptable provided that the cores will separate from each other.

0417. A binding tape in accordance with 0310 may be applied over the laid up cores, pairs, triads and quads and between the filler compound and outer sheath.

0418. Where a screen is specified for a core, pair, triad or quad, wrapping tape(s) may be applied to the core, pair, triad or quad prior to application of the screen. The braided screen is to be snug fit.

0419. If a collective screen is specified, a tape or extruded inner sheath is to be applied over the laid up cores, pairs, triads or quads prior to the application of the screen, as per the cable size quoted in the table of dimensions or by agreement between the user and manufacturer. A PTP tape may be applied over the screen if required for manufacturing purposes.

Screens

0420. Where required the braided tinned copper wire screen of the size quoted in the Tables of Dimensions is to be applied over the insulated core, twisted pair or laid up assembly of cores, pairs or triples. The braided screen is to be a snug fit. Joints in the braid wire are to be woven in or twisted so that the wire ends turn outwards. There are to be no joints in the completed braid. The number of spindles, ends and length of lay are to be such as to provide a braid with a minimum filling factor of 0.61 in order to achieve EMC/EMI test requirement, if required. The filling factor is defined as:-

$$\frac{mnd}{2\pi D} \left[1 + \frac{\pi^2 D^2}{L^2} \right]^{1/2}$$

Where

D = mean diameter of braid (mm)
d = diameter of braiding wire (mm)
L = length of lay of braiding wire (mm)
m = total number of spindles
n = total number of ends per spindle

0421. The filling factor shall be 0.70 where the coverage percentage of more than 90% is required to meet the EMI/EMC requirements. The coverage % is defined as:-

$C\% = (2f - f^2) \times 100$ Where f is the filling factor so obtained above.

Wherever it is not specified and where there is no agreement between the user and manufacturer, the screen wire nominal diameter shall be 0.2 mm for cable cores with diameter under screen up to 15 mm and shall be 0.3 mm for cable cores with diameter under screen above 15 mm. This is required to be separately indicated to the user.

Screen Insulation

0422. Adequate screen insulation is required for individual screened cables to avoid mutual abrasion of screens when the cable is subjected to the rigorous of installation and other handling hazards. To achieve this extruded or tape wrap materials may be used, but these should not contribute to the halogen content, toxicity level and flame retardancy of the cable. Tape wrapping is to be applied by one of the following methods:-

- (a) One tape with a Nominal thickness of 0.020 mm applied helically in opposite directions with a min. of 50% overlap.
- (b) Two tapes applied helically with a suitable overlap.
- (c) The width of the tape and angle of application is to be such as to minimize wrinkles and creases when the cable is bent.

Sheathing

0423. The sheath used in the construction of cables covered in this Document is to be fire retardant low smoke non toxic halogen free Electron Beam cross linked special polyolefin compound in accordance with test requirement mentioned in the specification. The thickness at any point should not be less than 80% of the nominal value. If the diameter over the laid up and filled cores is such that the addition of the specified sheath thickness will bring the cables above the specified maximum, the sheath thickness may be reduced by up to a maximum of 0.25 mm. The outer surface of the sheath is to be reasonably uniform and substantially circular so as not to degrade cable glanding.

Sheath Protection

0424. For multi core cables, at the request of the user annealed tinned copper wire or galavnised steel wire braid over the sheath may be provided for protection. The copper braid wire diameters are given in table of dimensions. The steel braid wire nominal diameter shall be 0.3 mm. In case the user desires the annealed tinned copper wire protective braid may be provided below the sheath. The option of sheath protection over/ under the sheath is to be decided in consultation with the user.

Tolerances

0425. Except where otherwise stated the overall diameter of the cable is not to depart from the value specified in the Tables of Dimensions by more than the appropriate tolerance given in Tables 4.1 below.

OVERALL DIAMETER OF CABLE		TOLERANCE	
Above mm	Upto and including mm	Minus mm	Plus mm
-	5	0.4	0.6
5	10	0.5	0.7
10	15	0.6	0.8
15	20	0.7	0.9
20	30	0.9	1.1
30	40	1.1	1.3
40	50	1.3	1.5
50	60	1.5	1.7
60	70	1.7	1.9
70	80	1.9	2.1
80	90	2.1	2.3
90	100	2.3	2.5

Table 4.1

0426. In addition all cables are to pass through a gauge 150 mm long with an internal diameter 0.25 mm greater than that obtained by the addition of the plus tolerance to the specified overall diameter of the cable as stated in the Tables of dimensions.

MARKING

Identification of Manufacturer

0427. Each cable is to be marked throughout its length with the year of manufacture printed on the cable sheath. The printing is to be well defined and permanent. In addition to, following be marked at an interval not exceeding 1 metre:-

- (a) Manufacturer's name or trade name.
- (b) Reference standard EE-50-13
- (c) Patt. Number provided by Navy and Size
- (d) "HR 120" "EBXL"
- (d) Voltage rating

0428 In addition, a coloured marker thread based on year of manufacture and gland size may be selected in accordance with the following tables:-

Year of manufacture	Colour of marker thread
2012	Green/Black
2013	Green/White
2014	Green/Red
2015	Black/White
2016	Black/Red
2017	White/Red
2018	Yellow/Blue

GLAND SIZE

GLAND SIZE	Maximum Diameter of cable (mm)
M10	10
M16	16
M25	25
M35	35
M50	50
M70	70

SECTION- V

INSPECTION AND TESTING

Criteria of Acceptance

501. All samples shall successfully pass all the specified tests for proving conformity with the requirements of this specification. The EBXL cables manufactured as per the material specifications and process contained in this Document will be subjected to the test schedule, necessary to check for any change in the ageing characteristics of component material used in manufacturing the EBXL cables of a complete cable and to confirm the performance of a complete cable.

Test Schedule

0502. Five types of tests are specified as follows and the tests appropriate to each type are listed in Table Annex-'C'.

- (a) **Routine Tests.** These tests are to be carried out by the manufacturer on all finished cable lengths to ensure consistency of the product.
- (b) **Acceptance Tests.** The tests are to be carried out on samples taken from a lot for the purpose of acceptance of the lot. Acceptance tests shall not be carried out on a particular size from the lot on which type tests have already been conducted.
- (c) **Type Tests.** These tests shall be carried out on all type of cables to prove conformity with the requirements of specification, general quality / design features of the cable. Complete type tests shall be carried out as per the schedule mentioned in Annex 'C'. The Type test once carried out for a particular type of cable shall remain valid for maximum period of three years. In case of any change in material or, design changes of the cable, complete type tests shall be repeated.

The sampling plan for the type tests is given below:

Sl.No.	Cable group	Cable to be tested
(a)	Taped insulated Single and multi core cables	≥ 25 Sq.mm
(b)	Extruded insulated Single and multi core cables	≤ 4 Sq.mm 7 core ≤ 4 Sq.mm 14 core
(c)	Multi pair cables, screened	≤ 1.0 Sq.mm, any item between 7 to 19 core
(d)	Multi core screened cables	≤ 1.0 Sq.mm, any item between 7 to 18 core

(d) **Product Development Tests.** These tests shall be carried out on the insulation and sheath materials to establish safe working temperature, life and Infra Red Spectrograph of cable materials.

(e) **Optional Tests.** Special Tests to be carried out when required by agreement between the user and the manufacturer.

0503. The permissible limit of each test is specified in the scheduled tests at Annex 'D' along with the test procedure and duration of the test. A brief description of tests which are carried out on compounding material/insulating material of cables and on the completed cables are enumerated in the succeeding paragraphs.

Tests on Cable Components

0504. The following tests are to be carried out on the cable components prior to or during manufacture as applicable.

(a) **Voltage Breakdown Tests on Compound.** Samples of material are to be conditioned in a loose unrolled state at $95 \pm 5\%$ Relative Humidity and $20^{\circ}\text{C} \pm 5^{\circ}\text{C}$ for not less than 96 hours. After removal from the conditioning chamber, the breakdown voltage is to be determined after 1 minute and within 4 minutes of removal. Breakdown which are due to 'flash-overs' are to be ignored. The waveform and frequency of the alternating voltage is to be as in BS 419 and is to be raised from zero to breakdown at $30 \text{ KV} \pm 5\text{KV}$ per minute.

Test Method. The apparatus used is to be generally similar to that described in BS 419 except that the upper electrodes are to have a nominal diameter of 6.35 mm and the lower electrodes 3.8 mm, the edge of the test ends being rounded to radius of 0.8 mm. The lower electrodes may be slightly recessed into the wall of the lower block of the apparatus so that they may just be covered with insulating oil, for the purpose of minimizing flash-over.

(b) **Extruded Insulated Cores Spark Test.** All extruded insulated cores are to be Spark Tested in accordance with BS 5099. The test is to be carried out after E-Beam cross-linking. The test voltage is to be as follows:

<u>Thickness of Insulation</u>	<u>Voltage ac (rms)</u>
Up to 0.35 mm	3000 V
0.35 mm – 0.50 mm	4000 V
Greater than 0.50 mm	6000V

Test on components from completed cable

Tensile strength & Elongation at Break of core

0505. Tensile strength and Elongation at break tests before and after accelerated ageing in air and oil as specified in Annex. 'D' to the Document.

Sheath

0506. Tests enumerated in the succeeding paragraphs are to be carried out on the samples taken from the sheath of completed cable which.

Heat Shock Test

0507. This test is conducted on sheath to determine the effect of elevated temperature on surface finish. The test specifications should conform to Annex 'D' to the Document.

Hot Set Test

0508. To confirm that the sheath has been adequately cross-linked the Hot Set Test to be carried out and should comply with Annex D of this Document.

Screen Solderability test.

0509. To confirm that the screen will "Wet" easily without damage to other cable components, the solderability test as specified in IEC-68-2-20, Method -I is to be carried out.

Tear Resistance

0510. The Tear Resistance Test is to be carried out in compliance with Def Stan 61-12 Part 31(1) Cl. 9.2 and is to comply with clause 0313 Table 3.1 of this Document.

Acid Gas Test/Corrosivity of Combustion gases.

0511. The Acid Gas Test in accordance with IEC 754-2 is to be carried out and is to comply with Table 3.1.

Thermal Endurance

0512. The thermal endurance profile of the overall sheathing material is to be determined as in BS 5691 Part 1 Clauses 11.2 and 11.3. or IEC-216. The end points to be used are to be 50% ultimate elongation of the sheathing material after continuous ageing as in BS 5691 Part 1 Clause 8b. or IEC-811-1-1 Cl. 9.0

0513. The Arrhenius plots so obtained are to form part of the submission to DQA (N) when seeking approval for the sheathing material. It will be required to show that the

sheathing material has a life exceeding 40000 hours at a continuous temperature of 120°C.

Accelerated Ageing

0514. Samples are to be prepared and tested as per IEC 811-1-1 Cl 8 and the results should comply with Annex 'D' of the Document.

Critical Oxygen Index

0515. Samples are to be tested to the requirements of ASTM-D-2863.

Temperature Index

0516. Samples are to be tested to the requirements of ASTM-D-2863, except that the test specimen of 6.5 mm width is to be a 70 mm to 150 mm length. Where necessary a thin wire support may be used.

Toxicity Index

0517. Samples of sheathing material to be tested in accordance with Annex 'D' to this Document.

Halogen Content

0518. The sheathing material is to be tested for potential acid Gas evolution by using the Lassaing test (sodium fusion test). A 0.25 gm (+/-100 mg) sample (approx. 25 mm length) of sheathing material, stripped from the test sample, and cut into small pieces, is to be subjected to the test. The test is to show negative result for fluorine, chlorine, bromine and iodine.

Smoke Density Rating

0519. Samples are to be prepared and tested as per ASTM-D-2843 and the results should comply with Annex 'D' of this Document.

Test on completed cables

0520. The tests on completed cables be carried out in accordance with the relevant specifications and the test results should comply with annex. 'D' to this Document. Brief on the tests is enumerated in the succeeding paragraphs.

Visual Examination

0521. Samples from completed cables are to be subjected to a visual examination to ensure that the cable is in full compliance with the constructional requirements specified and, notwithstanding the results of any of the specified performance tests, that it is free from following defects:-

- (a) The conductor wires are free from dis-colouration. Dulling or slight 'blueing' of the wires may be ignored.
- (b) The insulation is readily removable from the conductor.
- (c) Screens are to be a close but not tight fit on the core or assembly of cores.
- (d) The sheaths are homogeneous and capable of easy removal, without damage to any underlying component, except that adhesion to a tape may be ignored.
- (e) Marking on the cable surface.

Thermal Ageing /Ageing in Air Oven

0522. Samples are to be prepared and tested as per IEC 811-1-1 Cl 8 and the results should comply with Annex 'D' of this Document.

Flammability test

0523. Flammability test shall be carried out in accordance with IEC 332-1 and -3 for 4 sq. mm and above size of cables.

Electrical Tests

0524. All completed cables are to be tested dry at the contractor's works except that single core sheathed cables are to be tested in water in which the period of immersion is to be not less than 12 hours.

Fire survival Test

0525. The fire survival test shall be carried out at flame temperature 750°C +/-40°C and the duration of the flame application is to be 3 hours. The cable is to be re-energised not less than 12 hours after application of flame. The test be carried out as per BS 6387 at 250/ 440V. After the fire test, there should be no sign of severe kinking of the conductor.

Voltage Withstand Test

0526. The Voltage Withstand Test is to be carried out at ambient temperature with alternating voltage of sine wave form at any frequency between 40 and 60 Hz.

0527. The voltage is to be increased gradually and maintained steady at the full value for 5 minutes between each conductor and either water, screen or other cores which are to be earthed as applicable. For multi-core cables, grouping of cores may be adopted to reduce the number of tests. The cable is to comply with the requirements as specified in the Tables of Dimensions (Annex-'E')

Insulation Resistance Tests

0528. The Insulation Resistance Test is to be carried out immediately after the completion of the Voltage Withstand Test. The measurement is to be made at 500V DC $\pm$ 10% after 1 minute electrification. The apparent change in insulation resistance during electrification is to be steady. The results are to be corrected to Mega ohms for a 1 km length @ 20 degree C using the appropriate correction constants. The cable is to comply with the requirements as specified in the Tables of Dimensions (Annex-'E').

Conductor Resistance

0529. The conductor resistance of each conductor is to be measured at room temperature and corrected to 20⁰ C by the constants specified in IS8130/ BS 6360/ IEC 60228. The corrected resistance is not to exceed the values stated in the Tables of Dimensions.

Screen Insulation Resistance

0530. Insulation resistance between screened assemblies (cores, pairs or triples) is to be determined at 500 V DC +10% after 1 minute electrification. The result corrected for a length of 1 km of cable is to be not less than the value specified in the table of dimensions.

Capacitance Unbalance test

0531. The mutual capacitance unbalance is to be determined on all multi triple cables. The test is to be carried out at 1000 Hz between each combination of two cores in any one triple. The other core, cores and screen if applicable are to be connected together and to earth. The maximum difference between the capacitances measured between any two of the three cores of a triple is not to exceed 6.5 nF/km.

Dimensions

0532. The dimensions of the cable samples are to be measured in accordance with BS 6469 Clause 2.1 or IEC-811-1-1 for compliance with the relevant information contained in the Tables of Dimensions unless otherwise stated in the construction details.

Bend Tests

0533. A sample of the completed cable, with open ends, is to be bent round a test cylinder to make one complete turn. It is then to be unwound and the process repeated in the opposite direction. This cycle of operations is to be carried out 3 times. The diameter of the test cylinder is to be equal to 12 times the specified minimum cable diameter.

0534. The bending is to be carried out at a reasonably uniform speed so that each half cycle takes between 15 and 30 seconds, and at normal ambient temperature.

0535. On completion of bending, the cable is to be opened up and carefully examined. No component is to have sustained any damage or displacement and no conductor strands are to be broken. .

Resistance to Fluids

0536. Samples are to be tested in accordance with the schedule of tests Annex- 'D' & requirement as below.

Fluids	Test Temp	Duration	Requirement	Test Method
<u>Mineral OIL</u> <u>ASTM Oil No. 2</u>	100± 2°C	24 hrs	No crack or other deterioration in sheath. No Electrical breakdown	IEC-811-2-1
<u>Cleaner (Detergent)</u>	Room temp	24 hrs	No crack or other deterioration in sheath. No Electrical breakdown	IEC-811-2-1
<u>WATER</u> Deionized	Room temp	24 hrs	-do-	IEC-811-2-1
Deionized with 3 ¹ / ₂ NaCl	Room temp	24 hrs	-do-	IEC-811-2-1

TABLE 5.1. TEST FLUIDS, TEMPERATURES AND LIMITS

Note 1 Compliance with this test provides no guarantee that the sheath is suitable for continuous use in the above fluids.

Ozone Resistance

0537. Sample is to be tested in accordance with IS: 10810 Part 13. The ozone concentration is to be 80 to 100 ppm and the duration of exposure 5 days.

Cold Elongation

0538. This test is to be conducted on sheath after conditioning the sample at -20°C for 4 hours. Samples are to be tested in accordance with IEC-811-1-4 cl. 8.4 or MIL-C-24640A para 4.7.5 Tests at low temperature, elongation test at low temperature for sheaths, except that the test temperature is to be within the limits -20°C.

Spark Test on Collectively screened cable

0539. The collectively screened cables are to be tested for sheath spark to determine sheath extrusion continuity and free from micro holes etc. The test should comply with the specifications mentioned at Annex. 'D' to the EED.

Insulation Resistance Constant

0540. This test is to be carried out for outer sheath at room temperature on a 5 metre sample of collectively screened cable only which has no tape over the screen to the requirements of Table-3.1

Acceptance Test Limits

0541. Acceptance Test limits as specified in table 3.1 and Tests schedule should be met.

Fire Survival Tests

0542. Resistance to Fire with Mechanical shock: The fire survival test shall be carried out at flame temperature 750°C +/-40°C and the duration of the flame application is to be 15 Min with mechanical shock. The test be carried out as per BS 6387 at 250/440 V.

0543. Resistance to Fire with Water: The fire survival test shall be carried out at flame temperature 650°C +/-40°C and the duration of the flame application is to be 15 Min plus 15 min with water spray. The test be carried out as per BS 6387 at 250/440 V.

SECTION VI

QUALITY APPROVAL PROCEDURES

Definitions

0601. The Qualification Approval Authority which authorises the issue of Qualification approval Certificates in connection with this EE-50-13.

0602. DQA (N) is the Quality Assurance Authority and to be nominated in the relevant contract schedule (for production purposes).

General Requirements

0603. Qualification Approval is accepted as demonstrating that the manufacturer is capable of making the cable covered by the Qualification Approval Certificate.

0604. The manufacturer is to be currently on the List of Assessed Contractors and is to demonstrate, by virtue of having been assessed as complying with the requirements of relevant cable specifications in vogue. Further the manufacturer should have exercised the control at his works over materials, manufacturing processes and product testing in a way that the cables in production will comply in all respects with the requirements of this EE-50-13.

0605. Any change in the manufacturing specification for a cable, or cable component, for which a Qualification Approval certificate has been issued is to be notified to the Purchasing Authority with a copy to the MOD Quality Assurance Representative DQA (N).

0606. The manufacturer may be required, at the discretion of the Qualification Approval Authority, to carry out repeat Qualification Approval tests and/or supply evidence in the form of results of tests on cable samples.

0607. The manufacturer is not to release any cable which embodies changes described in Clause 0605 above without prior written consent of the IHQ MOD (Navy)/DEE or DQA (N).

Qualification Approval Testing

0608. Application to submit a cable for Qualification Approval is to be made to DQA (N) New Delhi, supported by the following information:

- (a) Proposed place of manufacture, and date for commencement of tests.
- (b) Manufacturer's registration number on the List of Assessed Contractors.

(c) Manufacturer's proposals regarding the place at which tests are to be performed.

(d) Details of the general composition of the core and screen insulation, fillers, binders, wrapping tapes and sheathing materials.

0609. The full cost of Qualification Approval testing are to be met by the manufacturer. For Qualification Approval purposes the manufacturer is to make a representative cable length that is sufficient to carry out all the tests specified in Annexure 'C'.

0610. The manufacturer may apply for Qualification Approval for any of the cable groups listed in this EE-50-13.

Test Reports

0611. Upon completion of all the scheduled tests, a report similar to that in Annex 'A', giving the results is to be submitted to the Qualification Approval authority. All reports are to be certified by a responsible person on behalf of the manufacturer and countersigned by the DQA (N).

Qualification Approval Certificates

0612. Qualification Approval Certificates will be issued by the Qualification Approval Authority provided the specified requirements are met in full. The Approval certificate number allocated is to be quoted in all correspondence.

0613. When a manufacturer makes a further submission incorporating modification in accordance with Clause 0605 and the results are satisfactory, a new Qualification Approval certificate will be issued and the original certificate cancelled.

0614. Qualification Approval Certificates will be reviewed periodically. If a manufacturer has a continuous break in production of any particular cable for 3 years or more, he may be required to obtain re-approval.

0615. Testing for Qualification approval purposes will be carried out on samples of any size as decided by IHQ MOD (N)/DEE and DQA (N).

SECTION VII

QUALITY ASSURANCE OF PRODUCTS

Procedure

0701. The manufacturer is to ensure Quality of Conformance to the specified requirements by the following method:-

- (a) Inspection of materials and cable in the course of manufacture to ensure that the requirements of this Document are met and the necessary standard of workmanship and process control is being maintained.
- (b) Subjection of cable and representative samples to the specified tests.

Tests and Sampling Levels on Completed Cables

0702. All cables are to be subjected to the Routine tests specified in Section 5. On satisfactory completion of Routine tests random samples are to be taken during cutting for dispatch and subjected to Acceptance Test. One sample of each size will be required for acceptance tests.

0703. The Quality Assurance Authority may at intervals select samples of cables for Quality audit or check testing in their own laboratory. Samples are to be selected by the supervising representative of the Quality Assurance Authority. The cost of the samples is to be met by the manufacturer

SECTION VIII

Packing and Labelling

0801. Immediately after final testing the cable ends are to be effectively sealed against ingress of moisture. Unless otherwise ordered, the cable is to be supplied in the lengths stated in the Table of Dimensions and is to be supplied on reels or drums as stated in the order or contract.

Supply on Reels

0802. Cables 15 mm or less overall diameter and not exceeding a mass of 25 Kg may be supplied on plywood reels or hardboard reels constructed from standard hardboard complying with the requirements of BS 1142 part 2. Both flanges and the barrel are to be of these materials. The barrel diameter is not to be less than 20 times the maximum specified diameter of the cables.

0803. Cable ends should be effectively sealed against ingress of moisture and the ends must be fixed on the drum surface (flange) to avoid becoming free during handling, stowage and transport.

0804. The cables be supplied on suitably sized Wooden/ Steel Drums based on size and weight of standard length of cable. Cables procured for stocking for period more than five years generally be supplied on steel drums.

0805. A suitable protective wrapping is to be used around the outside of the cable after winding on reels with sufficient overlap to retard the ingress of liquids.

0806. A weatherproof label is to be attached to the reel permanently and clearly marked with details quoted in contract and must include the following information:-

- (a) Tare weight in kg.
- (b) Length of cable in metres
- (c) Year of cable sheath manufacture
- (d) Contract number and date
- (e) Name of cable manufacture
- (f) Nominal cross-sectional area of the conductor.
- (g) Number of cores.
- (h) Direction of rotation of drum (by means of arrow)
- (j) Patt. number

Supply on Drums

0807. When the cables are to be supplied on drums, the drum shall comply with specifications mentioned in IS. 10418. Further the labelling information detailed in clause 0806 is to be permanently and clearly marked on the flange of each drum, either by stenciling or by stamping the details on a suitable metal plate nailed to the drum. The plate is to have a minimum thickness of 3 mm.

TECHNICAL PARTICULARS OF ELECTRICAL CABLE TYPE

Sheet No.	<u>General Description:-</u>	
	Specification: EE- 50-13 (Rev 1)	
End Application:-		
QAP Reference No. RCPL/QAP/26/07, Rev.00		
I. <u>Technical Particulars</u>		
(a) Size/Parameter i.e. core x CSA (Area of cross-section)		
(b) Voltage Grade (Volts)		
(c) Current rating(Amps)		
(d) Gland size		
II. <u>Conductor</u>		
(a) Material		
(b) Class of Conductor		
(c) Nominal Cross Sectional Area(Sq.mm)		
(d) No of wires/Nominal dia of wire		
(e) Max DC Resistance at 20 ⁰ C (/Km)		
(f) Nominal Conductor dia		
III. <u>Insulation</u>		
(a) Material		
(b) Nominal Radial thickness		
(c) Wrapping tape		
(d) Min Insulation resistance (M ~km)		
(e) Nominal Core diameter		
(f) Core identification		
(g) Individual Screening : i. Material ii. Nominal dia of wire iii. Min filling factor & Min Coverage		
(h) Nom. diameter over screened core		
(I) Individual Screen Insulation: i. No. & Overlap of tapes ii. Tape thickness iii. Dia over insulated screens		

Note: All dimensions are in mm

TECHNICAL PARTICULARS OF ELECTRICAL CABLE TYPE

IV. Laying	
(a) Nom. dia over laid up cores	
(b) Filling of core interstice	
V. Inner sheath / Wrapping tape	
(a) Min thickness of inner sheath	
(b) Nom. Thickness of Wrapping tape	
(c) Nom.dia over inner sheath / wrapping tape	
VI. Collective Screen	
(a) i. Material ii. Nominal dia of wire iii. Min filling factor & Min Coverage	
(b) Dia over collective screen	
(c) Manufacturing year	
(d) Colour of the Marker thread	
VII. <u>Sheath</u>	
(a) Material	
(b) Nom. Radial thickness of Sheath	
(c) Identification on Sheath	
(d) Nom. dia over Sheath	
(e) Overall Braiding: i. Material ii. Dia of wire (mm) iii. Min Coverage %	
(f) Overall diameter Min / Max (mm)	
(g) Approx. weight of the cable (Kg/Km)	
(h) Packing: Standard length on the Drum.	
Tolerance	
End sealing	

Note: All dimensions are in mm

STANDARD FORM OF REPORT FOR QUALIFICATION APPROVAL
PRODUCTION QUALITY AND PRODUCTION ROUTINE TESTS

Cable Description : ELECTRIC CABLES, FIRE SURVIVAL, HIGH TEMPERATURE ZONES AND FIRE RETARDANT HALOGEN FREE SHEATHED ELECTRON BEAM IRRADIATED CROSSLINKED CABLE

MANUFACTURER'S NAME AND ADDRESS:-

Order No:

Patt No.

Type:

Length:

MATERIALS USED IN CONSTRUCTION

(EE- 50-13 Clause(s) in brackets)

Component		Material Used
Conductor	(0301)	
Separator Tape	(0305)	
Cores	(0306)	
Insulation	(0306)	
Thickness (Extruded / Taped)		
Mica tape	(0306)	
Braid	(0308)	
Lacquer	(0309)	
Fillers, Binders, Wrapping Tapes	(0310) (0311)	
Electrical Screen	(0312)	
Sheath	(0313)	

CONSTRUCTION OF CABLES
(EE- 50-13 Clause(s) in brackets)

Operation	Specified	Actual
Stranding (0403) a) Size of conductor b) Diameter of conductor c) Direction of lay		
Insulation Thickness (Extruded /Taped) (0407) Min Mean No. of tapes Diameter over insulation		
Glass fiber braiding / tape (0407)(0408) (0409) a) Lay angle b) Lacquer c) Diameter over braid / tape & lacquer		
Wrapping Tapes (0310)		
Core Identification (0412)		
Laying up of Cores (0414) a) Formation b) Lay c) Direction of Lay		
Screening (0420 -0421) a) Size of wire b) Filling factor c) Diameter over screen		
Screen Insulation (0422) a) Number of tapes b) Overlap of tapes c) Diameter over insulated screens		
Sheathing (0423) Thickness Min Mean		
Overall Diameter Min (0425-0426) Max		
Marking (0427) a) Manufacturer b) Year c) Ref Spec no. EED 50 11 d) Part No. / Patt. No.		
Additional Marking (0427)		

TESTS

TESTS		Specified	Actual
Spark test on insulated cores	(0504)b		
<u>Type –E insulation</u>			
Tensile strength	(0505)		
Elongation at break (Before accelerated aging)			
Elongation at break (After accelerated aging)			
<u>Type –T insulation</u>			
Breakdown voltage	(0504)a		
Tensile strength			
Thickness	Min. & Max.		
<u>Physical Tests on Sheath</u>			
Tensile strength	(0505)		
Elongation at break	(0505)		
Tear resistance	(0510)		
Cold Elongation	(0538)		
Heat shock test	(0507)		
Hot set test	(0508)		
Flammability of core	(0523)		
Conductor resistance	(0529)		
Voltage test	(0526)(0527)		
Insulation resistance test (Cores)	(0528)		
(Between screens)	(0530)		
Capacitance unbalance	(0531)		
Spark test on collectively screened cables	(0539)		
Heat endurance test			
Bending	(0533-0535)		
Flammability	(0523)		
Fire survival	(0525)		
Optional Fire Survival tests	(0542-0543)		
Visual Examination	(0521)		
Screen Solder ability	(0509)		
Thermal Ageing	(0523)		

GUIDELINES FOR SELECTION OF CABLES (Def Stan 02-512 Part 1 to 15)

1. Def Stan 02-512 Part 1 to 15 gives the technical data on electrical cables and associated items intended for use in the design and approval of electrical system and equipment for Naval ships. These design data have been reworked based on the EBXL technology and are specific to the Electron Beam Cross linked irradiated cables only. The data have been presented in a tabular manner at the table of dimensions for easy reference of users.

Arrangement of Tables.

2. To facilitate reference to data, uniformity of presentation has been preserved throughout. Columns read in the same general order from left to right, and column entries show cables in sequence of cross sectional areas, and number of cores. Tabular presentation has therefore been made even where all cables in a range have many identical characteristics.

3. Cables which fulfill more than one service are shown in tables for each function, e.g.; two and three core cables which are used for power or control circuits.

Service of Cables

4. The CONTENTS and Table Headings have been annotated to indicate the service for which cables are mainly used. This is not exhaustive, and does not preclude use of cables for other services.

Dimensions and Weight

5. Dimensions and weight values are nominal unless otherwise stated, e.g.; overall diameter of cable is given as maximum.

6. Diameters over insulated conductors, pairs etc include the thickness of tapes. Diameters under screens of collectively screened cables are maxima deduced from OD of cables.

Resistance

7. Resistance values given are maxima per conductor at 20 degree C.

Voltage Rating

8. Voltage rating given throughout is nominal RMS value, and nominal voltage of AC circuits for which cables are selected should not exceed the stated rating. For dc circuits nominal voltage should not exceed the peak value of the stated ac voltage.

Volt Drop

9. Reactance and ac resistance values and the ROUTE LENGTH of cable in meters which will give a volt drop of 1 volt at the current rating and power factor given have been included in the tables of cables designed for use on single phase and three phase power circuits and have been calculated for 60 Hz systems. When the cables are used on 400 Hz systems the figures given in the Table of Dimensions.

10. ONE VOLT DROP means a difference of 1V in the voltage measured across lines at the supply and load extremities of the cable.

11. Volt drop data may also be applied to adjacent cores of multi-core cables of equivalent CSA, subject to pro-rate adjustment for differences in current.

12. For current loading of less than maximum, lengths for 1V drop may be taken as inversely proportional to current.

Current Rating

13. Current rating given throughout is maximum continuous rating for ac (RMS value) and dc circuits in an ambient temperature of 50 degree C unless otherwise stated :-

A(1) is the rating for a single core cable.

For multi-core cables with all cores equally loaded

A(2) is the rating when used on 60 Hz systems

A(3) is the rating when used on 400 Hz systems.

14. For varying numbers of cores equally loaded the following correction factors may be applied to the equivalent of the single core rating.

No of cores equally loaded	2	3 & 4	5-8	9-15	16-25	26-44
Correction factor %	80	70	60	50	45	40

15. For ambient temperatures other than 50 degree C the following correction factors may be applied (to any given current rating):-

Ambient Temp degree C	55	60	65	70	75	80	85	90	95	100
Correction factor %	95	90	85	80	74	67	60	52	43	30

Core Identity and Layering

16. For identity of cores refer clause no. 0412 & 0413 and for layering refer table of dimensions.

Selection criteria of Electron Beam Irradiated cross linked cables:

17. The E-Beam cross linked technology provides high current carrying capacity improved mechanical and thermal properties for rugged applications. The cables are of compact design, have smaller core and sheath diameter and are lighter in weight as compared to the conventional LFH cables. The introduction of EBXL cables there fore offers significant savings in terms of weight and cost.

18. In order to ensure that the standard gland sizes in use for LFH cables are retained in case of EBXL induction on board new construction / retrofitment, the internationally accepted LFH cable sizes have been mapped on to the EBXL cable sizes. Comparative tables of EBXL cables with reference to standard LFH cables has been formulated. Same is placed at Annexure "F" and may be used as reference while ordering EBXL cables. However, one must while selecting the cables should also make a reference to the maximum current requirement of the system.

ABBREVIATIONS USED IN TABLES

CABLE DATA

A(1)	Current rating, single core cables
A(2)	Current rating, multi-core cables , 60 Hz systems
A(3)	Current rating multi-core cables, 400 Hz systems
C	Number of cores in cable
CSA	Cross sectional area of conductor
D	Diameter of wires and conductors
DUS	Diameter under screen of collectively screened cables
DSW	Diameter of screen wire
EBXL	Electron Beam cross linked
LD	Length of cable on drum (or reel)
OD	Overall diameter of cable
ODFS	Overall diameter of filled screen (Pressure Tight cables).
ODIC	Overall diameter of insulated conductor
ODSI	Overall diameter of screen insulation
ODTP	Overall diameter of twisted pair
P	Number of pairs in cable
PF	Power factor
Ph	Phase
ATC	Annealed Tinned Copper
R	Resistance
Rac	ac resistance
Tow	Tinned copper wire

UPF	Unity power factor
V	Maximum ac voltage rating
W	Weight of cable
X	Reactance

CORE IDENTITY

BK	Black
F	Filler
R	Red
U	Uncoloured

TEST SCHEDULE

Test	Type Test / Qualification Approval	Acceptance Test	Routine Test	Product Development Test (Type Test+ Endurance Test)
I. INSULATION / CORE (COMPONENT)				
Voltage Breakdown of insulation tape	√			√
Tensile Strength of insulation tape	√			√
Thickness of insulation tape	√			√
Spark test of extruded core			√	√
Extruded Insulation	√	√		√
a) Tensile Strength				
b) Elongation				
c) After Ageing – Elongation Retention				
Flammability of core	√			√
II. Sheath				
Mechanical properties	√	√		√
Tensile Strength				
Elongation at break			√	
Ageing in Air Oven 150 Deg.C, 28 days,	√			√
Absolute Elongation				
Hot Oil Ageing – ASTM 2 100°C/24 hrs	√			√
Tensile Strength				
Elongation at break				
Hot set test	√			√
Tear Strength	√			√
Ozone Resistance (120 ± 2 hours exposure at 80 to 100 ppm at room temperature)	√			√
Resistance to fuel	√			√
Fuel oil – after immersion for 24 Hrs at 70 ± 2°C				
Oxygen Index	√	√		√
Temperature Index	√	√		√
Cold Elongation Test (-) 20 ± 3°C for 4 hours	√			√

Test	Type Test / Qualification Approval	Acceptance Test	Routine Test	Product Development Test (Type Test+ Endurance Test)
Acid gas generation i.e., Corrosivity of combustion gases Test sample restricted to 1 sample from each lot.	√	√		√
Toxicity of combustion gases	√	√		√
Smoke Density rating Or Smoke Index	√	√		√
Thermal Endurance test & duration of usability elongation at break absolute. If this test is conducted as PDT then repeat test as type test is not required	√	√		√ If this test has been conducted as per EED 50-12 then repetition of this test is not required.
Resistance to fluids. a) ASTM Oil No. 2 (100°C/100 hrs) b)) Diesel (Fuel Oil at 70 ±2°C for 24 hours)	√			√
Polymer Identification Test	√			√
Halogen Content.	√			√
Sheath Compatibility. Variation in T.S. & Elongation temp. 120°C for 7 days for CPE and 82°C for 7 days for PVC.	√			√
IR Constant (Sheath)	√			√
Pressure test - Indentation	√			√
Resistance to Ultra Violet (1000 Hrs.)				√
III. COMPLETED CABLES				
Construction, Dimension & Thickness	√	√	√	√
D.C Resistance of Conductor	√	√	√	√
Insulation Resistance Constant at 20°C for screened and unscreened cables	√	√		√
Voltage Test Core/Core, Core/Screen Core/water or Spark test for single core unscreened cable only.	√	√	√	√

Test	Type Test / Qualification Approval	Acceptance Test	Routine Test	Product Development Test (Type Test+ Endurance Test)
Flame Retardance (Flammability) (Single Cable)	√	√		√
Flame Retardance (Flammability) (Bunched Cable)	√			√
Capacitance unbalance Test for Pairs/triads/quads	√	√		√
Capacitance Test on pairs/triad/quad	√	√		√
Bend Test	√	√		
Insulation Resistance Constant for collectively screened cables outer sheath.	√	√		√
Spark test on collectively screened cable	√	√		√
Abrasion Resistance test				√
Fire Survival test		√		√
Visual Examination	√		√	√
Heat Endurance				√
IV. FIRE SURVIVAL TESTS				
Resistance to Fire with Mechanical shock				√
Resistance to Fire with Water				√
V. SCREEN				
Screen Solderability Test	√	√		√

VARIOUS TESTS, THEIR REQUIREMENTS & METHODS FOR ELECTRIC CABLES

Sl. No.	TESTS	CLASSIFICATION				REQUIREMENTS	TEST METHODS
		RT	AT	TT	PDT		
	INSULATION / CORE (COMPONENT)						
	Voltage Breakdown of insulation tape			✓	✓	Min. 28kV / mm	BS 419 Def. Stan. 02-527 (NES 527)
	Tensile Strength of insulation tape			✓	✓	Min. 16 N/mm	BS 419
	Spark test of extruded core	✓			✓	3000 V upto 0.35mm 4000 V 0.35mm – 0.50mm 6000 V greater than 0.50mm	BS 5099
	Extruded Insulation		✓	✓	✓		
	a) Tensile Strength					Min. 8.0 N / mm ²	BS 6469 /
	b) Elongation					Min. 150%	IEC 811-1-1 Cl 9 / IS 10810 Part 7
	c) After Ageing – Elongation Retention					Min. 75% (150°C for 168)	IEC 811-1-1 Cl 8 / IS 10810 Part 11
	Flammability of core			✓	✓	Charred portion shall not have reached within 50 mm of the lower edge of the top lamp	BS 4066 Part 1 / IEC 332 part 1
2.0	Sheath						
2.1	Mechanical properties Tensile Strength Elongation at break		✓	✓	✓	> 8.0 N/Sq.mm > 200 %	IEC 811-1-1 Cl 9 / IS 10810 Part 7 Def. Stan. 61-12 Part 31 Issue 2

Sl. No.	TESTS	CLASSIFICATION				REQUIREMENTS	TEST METHODS
		RT	AT	TT	PDT		
2.2	Ageing in Air Oven 150 Deg.C, 28 days, Tensile Strength			✓	✓	Variation ± 50%	IEC 811-1-1 Cl 8 / IS 10810 Part 11. After ageing for 28 days the median value of Tensile Strength is not to be less than 50% of the Tensile strength without ageing. The difference in the elongation at break after ageing for 14 days and 28 days respectively is not more than 20% of the elongation at break without ageing. After ageing for 28 days the median value of elongation at break is not to be less than 120%.
	Elongation at break					Variation ± 20%	
	Absolute Elongation					120%	
2.3	Hot Oil Ageing – ASTM 2 100°C/24 hrs			✓	✓		IEC 811-2-1, Cl.10
	Tensile Strength					Min. 50 % of Original	
	Elongation at break					Min. 50 % of Original	
2.4	Hot set test					200 ± 5° C	
	- Stress - Hot Set Elong. - Permanent Set					0.2 N/mm ² Max 175% Max 25 %	BS 6469 Cl. 3.3 / IS 10810 Part 30 / IEC- 811-2-1 Cl 9
2.5	Tear Strength			✓	✓	Min. 5 N/mm	Def Stan 61-12 Part 31/1 Cl. 9.2
2.6	Ozone Resistance (120 ± 2 hours exposure at 80 to 100 ppm at room temperature)			✓	✓	No cracks.	IS 10810 Part 13 / UIC 895 OR Cl.5.2.4
2.7	Resistance to fuel			✓	✓		IEC-811-2-1, Cl. 10
	Fuel oil – after immersion for 24 Hrs at 70 ± 2°C i) Tensile Strength ii) Elongation at break					Min. 50 % Retention Min. 50 % Retention	

Sl. No.	TESTS	CLASSIFICATION				REQUIREMENTS	TEST METHODS
		RT	AT	TT	PDT		
2.8	Oxygen Index		✓	✓	✓	≥ 29	ASTM D 2863
2.9	Temperature Index		✓	✓	✓	≥ 250 Deg. C	ASTM D 2863 & BICC Hand book
2.10	Cold Elongation Test (-) 20 ± 3°C for 4 hours			✓	✓	Min. 20%	IEC 811-1-4 Cl 8.4 or MIL -C-24640A
2.11	Acid gas generation i.e., Corrosivity of combustion gases Test sample restricted to 1 sample from each lot.		–	✓	✓	pH Index ≥ 4.3 , Electrolytic Conductivity ≤ 100 micro.S/ cm	IEC- 754-2
2.12	Toxicity of combustion gases			✓	✓	Toxicity Index ≤ 5	Gases as per UITP Part 2E 7 & test method as per NES-713
2.13	Smoke Density rating Or Smoke Index		✓	✓	✓	≤ 20 % Max. ≤30%	ASTM-D-2843 IEC-1034-2
2.14	Thermal Endurance test & duration of usability elongation at break absolute. If this test is conducted as PDT then repeat test as type test is not required.				✓	≥ 50% after 40000 hrs at 120 ± 2°C	IEC 216, Arrhenius Plot
2.15	Resistance to fluids. a) ASTM Oil No. 2 (100°C/100 hrs) b)) Diesel (Fuel Oil at 70 ±2°C for 24 hours)			✓	✓	a) The insulation shall show no cracks, or ruptures or splits b) No breakdown when tested for high voltage test.	The central 25 cms section of a 50 cms length of core with ends stripped 5 cms shall be exposed to the fluids.
2.16	Polymer Identification Test			✓	✓	IR spectrograph shall be generated by the manufacturer for identification of polymer. This should be certified by the Inspecting Authority for future reference.	ASTM-D 3677
2.17	Halogen Content.			✓	✓	Zero Halogen	Def Stan 61-12 Part 31(1) Cl. 9.8
2.18	Sheath Compatibility. Variation in T.S. & Elongation temp. 120°C for 7 days for CPE and 82°C for 7 days for PVC.			✓	✓	CPE & PVC tapes will be used by wrapping the same on the cable specimens & physical properties of EBXL sheath will be tested. The variation should not be more than+/- 10%.	

Sl. No.	TESTS	CLASSIFICATION				REQUIREMENTS	TEST METHODS
		RT	AT	TT	PDT		
2.19	IR Constant (Sheath)			✓	✓	Min. 0.1 MΩ - km	Def Stan 61-12 Part 31 issue 2
2.20	Pressure test - Indentation			✓	✓	Max.50%,at120± 3°C	BS 6469 Section 4.2.3
2.21	Resistance to Ultra Violet (1000 Hrs.)				✓	UV 55 ± 3°C, Humidity 40 ± 3	Def Stan 61-12 (Part 0) Clause 16.16 / ASTM G 53
	- Tensile Retention				Min. 50%		
	- Elong. Retention				Min. 50%		
3.0	COMPLETED CABLES						
3.1	Construction, Dimension & Thickness	✓	✓	✓	✓	Tables of dimension	IEC-811-1-1, Cl.No. 8/ IS-10810
3.2	D.C Resistance of Conductor	✓	✓	✓	✓	Tables of dimension	IS-10810 Part 5 / IEC 228/BS-6360
3.3	Insulation Resistance Constant at 20°C for screened and unscreened cables		✓	✓	✓	Tables of dimension	IS-10810 Part 43 / IEC-885-1
3.4	Voltage Test Core/Core, Core/Screen Core/water or Spark test for single core unscreened cable only.	✓	✓	✓	✓	Test Voltage as per tables of dimension, duration 5 minutes.	IS-10810 Part 45 / IEC-885-1
3.5	Flame Retardance (Flammability) (Single Cable)		✓	✓	✓	Charred portion shall not have reached within 50mm of the lower edge of the top clamp.	BS 4066 Part 1 / IEC 332 Part 1
3.6	Flame Retardance (Flammability) (Bunched Cable)			✓	✓	Max.2.5m above the bottom edge of the burner.	BS 4066 Part 3 / IEC 332 Part 3
3.7	Capacitance unbalance Test for triples		✓	✓	✓	The capacitance unbalance should not exceed 6.5 nF/KM	
3.8	Capacitance Test triples		✓	✓	✓	Values obtained should comply with Table of Dimensions.	
3.9	Bend Test		✓	✓	✓	No cracks should develop	Para 0534-0537 of EED-50-13.
3.10	Insulation Resistance Constant for collectively screened cables outer sheath.		✓	✓	✓	0.1M ohms KM	10810 part 43

Sl. No.	TESTS	CLASSIFICATION				REQUIREMENTS	TEST METHODS
		RT	AT	TT	PDT		
3.11	Spark test on collectively screened cable		✓	✓	✓	6 kV	BS-5099
3.12	Abrasion Resistance test				✓	Min. 75 Cycles	MIL-DTL-24640B
3.13	Fire Survival test				✓	Flame temperature 750±40°C for 3H Shall withstand test voltage of 250 / 440 Volts as per table of Dimensions. No sign of severe kinking.	BS 6387
3.14	Visual Examination	✓			✓	Conductor free from discolouration. Insulation readily removable. Screens – good fit readily cleaned. Sheath capable of easy removal. Legibility of core identification.	Cl. 0537 of Def. Stan. 02-527 (NES 527)
3.15	Heat Endurance				✓	Conductor temperature 125°C for 400H Ext. Ins.show no significant change in hardness (or) colour (or) serious degree of adhesion to cond. Taped ins. Capable of removing without stripping the base insulant from supporting fabric. No significant change in the legibility of identification No.(s) Conductor show no sign of bird caging of kinking.	Cl. 0539 to 0548 of Def. Stan. 02-527 (NES 527)
4.0 FIRE RESISTANCE TESTS							
4.1	Resistance to Fire with Mechanical shock		✓		✓	Bending radius 6D & is hit by MS rod at 30± 2 sec per one hit. Shall withstand test voltage of 250 / 440V as per Table of Dimensions. Flame temp. 750 ± 40°C for 15min with mechanical shock.	BS 6387

Sl. No.	TESTS	CLASSIFICATION				REQUIREMENTS	TEST METHODS
		RT	AT	TT	PDT		
4.2	Resistance to Fire with Water		√		√	Spray water at a pressure between 250 & 350 kpa. Shall withstand test voltage of 250 / 440V as per Table of Dimensions. Direct Flame at temp. $650 \pm 40^{\circ}\text{C}$ for 15min. plus 15 minutes with water spray.	BS 6387
5.0 SCREEN							
5.1	Screen Solderability Test			√	√	No damage to other components of the cable during soldering	IEC 68-2-20 Method 1

Note: Classification of inspection

- RT - Routine Test
- AT - Acceptance Test
- TT - Type Test
- PDT - Product Development Test

TABLES OF DIMENSIONS
PART C
TABLE C1 : SINGLE CORE CABLES

1	2			3		4	5	6	7	8	9	10		11	12	13
Stock No.	Conductor			Insulation System		Sheath thickness (Nom)	Overall Diameter (Nom)	Voltage test	Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable	Route length for 1V drop, 1 Phase	Route length for 1V drop, 3 Phase
	Nominal Area	No. of Wires (typical value) x Single Wire Dia Maximum	Diameter (Nominal)	Radial thickness (Nominal)	Diameter Over protective layer and Lacquer (Nom)							Voltage	Current			
	sq.mm	No/mm	mm	mm	mm							V	A	kg / km	m	m
5013-6808A1	0.75	24/0.21	1.12	0.5	2.52	1.0	5.0	2000	26.7	um	500	440	17.0	38	1.1	1.3
5013-6808	1	32/0.21	1.3	0.5	2.7	1.2	5.6	2000	20.0	100	500	440	23.0	41	0.82	0.95
5013-6809	1.5	30/0.26	1.55	0.5	2.95	1.2	5.9	2000	13.7	100	500	440	33.0	49	0.83	0.96
5013-6810	2.5	50/0.26	2	0.6	3.6	1.2	6.5	2000	8.21	100	500	440	43.0	65	1.07	1.23
5013-6811	4	56/0.31	2.5	0.6	4.1	1.2	7.0	2000	5.09	100	500	440	55.0	87	1.34	1.55
5013-6812	6	84/0.31	3.1	0.7	4.9	1.2	7.8	2000	3.39	100	500	440	79.0	116	1.41	1.62
5013-6813	10	80/0.41	4.1	0.7	6.1	1.2	9.0	2000	1.95	100	500	440	100	183	1.93	2.23
5013-6814	16	126/0.41	5.4	0.8	7.6	1.2	10.5	2000	1.24	100	500	440	140	247	2.17	2.50
5013-6815	25	189/0.41	6.8	0.8	9.0	1.2	11.9	3500	0.795	55	250	440	165	357	2.87	3.31
5013-6816	35	259/0.41	7.8	0.8	10.0	1.2	12.7	3500	0.565	45	250	440	200	446	3.33	3.85
5013-6817	50	378/0.41	9.5	0.8	11.7	1.2	14.7	3500	0.393	45	250	440	250	565	3.83	4.43
5013-6818	70	348/0.51	11.5	0.8	13.7	1.3	16.8	3500	0.277	40	250	440	315	820	4.32	4.98
5013-6819	95	456/0.51	12.9	1.0	15.7	1.4	19.1	3500	0.210	40	250	440	380	1053	4.72	5.45
5013-6820	120	570/0.51	14.8	1.0	17.6	1.5	21.5	3500	0.164	40	250	440	420	1526	5.47	6.31
5013-6821	150	741/0.51	16.8	1.0	19.6	1.5	23.5	3500	0.132	35	250	440	480	1615	5.94	6.86
5013-6822	185	893/0.51	18.5	1.0	21.3	1.6	25.4	3500	0.108	30	250	440	560	2120	6.23	7.19
5013-6823	240	1184/0.51	21.0	1.0	23.8	1.7	28.2	3500	0.0817	25	250	440	660	2586	6.98	8.06
5013-6824	300	1480/0.51	23.5	1.0	26.3	1.8	30.7	3500	0.0654	25	250	440	790	3196	7.29	8.41
5013-6825	400	1920/0.51	27.0	1.0	29.8	1.9	34.6	3500	0.0495	20	250	440	890	4153	8.55	9.87
5013-6826	500	1690/0.61	31.2	1.0	34.0	2.1	39.3	3500	0.0391	20	250	440	970	5194	9.93	11.46
5013-6827	630	2236/0.61	35.2	1.1	38.2	2.2	43.8	3500	0.0292	20	250	440	1180	6542	12.28	14.18

TABLES OF DIMENSIONS**PART C****TABLE C2: TWIN AND THREE CORE CABLES**

1	2	3			4		5	6	7	8	9		10	11	12	13		14	15	16
Stock No.	No. of Cores	Conductor			Insulation System		Diameter Over taped Laid up Cores (Nom)	Sheath thickness (Nominal)	Diameter Overall (Nominal)	Test Voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable	Route length for 1V drop, 1 Phase	Route length for 1V drop, 3 Phase
		Nominal Area	No. of Wires (typical value) x Single Wire Dia Max	Diameter (Nominal)	Radial thickness (Nominal)	Dia. Over Protective layer and Lacquer (Nom)					Test Voltage	Minimum Breakdown Voltage				Voltage	Current			
		Sq.mm	No/mm	mm	mm	mm	mm	mm	mm	V	V	V	Ohm	Meg.O hm	m	V	A	kg / km	m	m
5013-6828A1	2	0.75	24/0.21	1.1	0.5	2.5	5.2	1.2	8.2	2000	440	500	26.7	100	500	440	13.6	80	1.0	1.2
5013-6828	2	1.0	32/0.21	1.3	0.5	2.7	5.6	1.2	8.6	2000	440	500	20.0	100	500	440	18.5	90	1.02	1.18
5013-6829	2	1.5	30/0.26	1.55	0.5	2.95	6.1	1.2	9.0	2000	440	500	13.7	100	500	440	26.4	124	1.04	1.20
5013-6830	2	2.5	50/0.26	2.0	0.6	3.6	7.4	1.2	10.4	2000	440	500	8.21	100	500	440	34.4	168	1.33	1.54
5013-6831	2	4.0	56/0.31	2.5	0.6	4.1	8.4	1.2	11.5	2000	440	500	5.09	100	500	440	44.0	217	1.68	1.94
5013-6832	2	6.0	84/0.31	3.1	0.7	4.9	10.0	1.2	13.1	2000	440	500	3.39	100	500	440	63.2	296	1.76	2.03
5013-6833	2	10	80/0.41	4.1	0.7	5.9	12.0	1.3	16.2	2000	440	500	1.95	100	250	440	80.0	450	2.41	2.79
5013-6834	2	16	126/0.41	5.4	0.8	7.6	15.4	1.4	19.0	2000	440	500	1.24	100	250	440	112	663	2.71	3.13
5013-6835	2	25	189/0.41	6.8	0.8	9.0	18.2	1.5	22.2	3500	440	500	0.795	55	250	440	132	845	3.59	4.14
5013-6836	2	35	259/0.41	7.8	0.8	10.0	20.2	1.6	24.4	3500	440	500	0.565	45	250	440	160	1030	4.16	4.81
5013-6837	2	50	378/0.41	9.5	0.8	11.7	23.6	1.7	28.0	3500	440	500	0.393	45	250	440	200	1450	4.79	5.53
5013-6838	2	70	348/0.51	11.5	0.8	13.7	27.6	1.8	32.2	3500	440	500	0.277	40	250	440	252	2150	5.39	6.23
5013-6839	2	95	456/0.51	12.9	1.0	15.7	31.6	2.0	36.8	3500	440	500	0.210	40	250	440	304	2484	5.90	6.81

TABLES OF DIMENSIONS**PART C****TABLE C2: TWIN AND THREE CORE CABLES**

1	2	3			4		5	6	7	8	9		10	11	12	13		14	15	16
Stock No.	No. of Cores	Conductor			Insulation System		Diameter Over taped Laid up Cores (Nom)	Sheath thickness (Nominal)	Diameter Overall (Nominal)	Test Voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable	Route length for 1V drop, 1 Phase	Route length for 1V drop, 3 Phase
		Nominal Area	No. of Wires (typical value) x Single Wire Dia Max	Diameter (Nominal)	Radial thickness (Nominal)	Dia. Over Protective layer and Lacquer (Nom)					Test Voltage	Minimum Breakdown Voltage				Voltage	Current			
		Sq. mm	No./mm	mm	mm	mm	mm	mm	mm	V	V	V	Ohm	Meg. Ohm	m	V	A	kg / km	m	m
5013-6840A1	3	0.75	24/0.21	1.1	0.5	2.5	5.6	1.2	8.5	2000	440	500	26.7	100	500	440	11.9	95	1.2	1.4
5013-6840	3	1.0	32/0.21	1.3	0.5	2.7	6.0	1.2	9.1	2000	440	500	20.0	100	500	440	16.1	115	1.2	1.35
5013-6841	3	1.5	30/0.26	1.6	0.5	2.95	6.6	1.2	9.5	2000	440	500	13.7	100	500	440	23.1	138	1.2	1.37
5013-6842	3	2.5	50/0.26	2.0	0.6	3.6	8.0	1.2	11.1	2000	440	500	8.21	100	500	440	30.1	205	1.5	1.76
5013-6843	3	4.0	56/0.31	2.5	0.6	4.1	9.1	1.2	12.1	2000	440	500	5.09	100	500	440	38.5	260	1.9	2.22
5013-6844	3	6.0	84/0.31	3.1	0.7	4.9	10.8	1.2	13.9	2000	440	500	3.39	100	500	440	55.3	361	2.0	2.32
5013-6845	3	10	80/0.41	4.1	0.7	5.9	12.9	1.4	17.4	2000	440	500	1.95	100	250	440	70.0	590	2.8	3.19
5013-6846	3	16	126/0.41	5.4	0.8	7.6	16.6	1.4	20.1	2000	440	500	1.24	100	250	440	98.0	818	3.1	3.58
5013-6847	3	25	189/0.41	6.8	0.8	9.5	20.7	1.6	23.8	3500	440	500	0.795	55	250	440	115	1200	4.1	4.73
5013-6848	3	35	259/0.41	7.8	0.8	10.6	23.1	1.6	25.9	3500	440	500	0.565	45	250	440	140	1575	4.8	5.50
5013-6849	3	50	378/0.41	9.5	0.8	12.3	26.8	1.8	30.0	3500	440	500	0.393	45	250	440	175	1822	5.5	6.32
5013-6850	3	70	348/0.51	11.5	0.8	14.5	31.5	1.9	34.7	3500	440	500	0.277	40	250	440	220	2442	6.2	7.12
5013-6851	3	95	456/0.51	12.9	1.0	16.5	35.8	2.1	39.5	3500	440	500	0.210	40	250	440	266	3820	6.7	7.78
5013-6852	3	120	570/0.51	14.8	1.0	18.4	39.9	2.3	44.2	3500	440	500	0.164	40	250	440	294	4575	7.8	9.02
5013-6853	3	150	741/0.51	16.8	1.0	20.4	44.3	2.4	48.7	3500	440	500	0.132	35	250	440	336	5480	8.5	9.80
5013-6854	3	185	893/0.51	18.5	1.0	22.1	47.9	2.6	52.8	3500	440	500	0.108	30	250	440	392	6400	8.9	10.27
5013-6855	3	240	1184/0.51	21.0	1.0	24.6	53.3	2.8	58.7	3500	440	500	0.082	25	250	440	462	8788	10.0	11.52

TABLES OF DIMENSIONS**PART C****TABLE C3: MULTI CORE CABLES**

1	2	3			4		5	6	7	8	9	10		11	12	13	14		15
Stock No.	No. of Cores	Conductor			Insulation		No. of Cores per layer	Diameter Over taped laid up Cores (Nom)	Sheath thickness (Nom)	Overall Diameter (Nom)	Test voltage	Gas Flame Test		Max. Conductor Resi. Per km at 20°C	Min. Insulation Resi. Per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (typical value) x Single Wire Dia Max	Diameter (Nominal)	Radial thickness (Nominal)	Dia. Over Protective layer and Lacquer (Nom)						Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq. mm	No./mm	mm	mm	mm		mm	mm	mm	V	V	V	Ohm	Meg. Ohm	m	V	A	kg / km
5013-6856A1	7	10	80/0.41	4.1	0.9	6.3	F.7	21.4	1.8	25.8	2000	440	500	1.95	100	250	440	60	1540
5013-6856	7	16	126/0.41	5.4	0.9	7.6	F.7	25.8	1.8	30.2	2000	440	500	1.24	100	250	440	84	1920
5013-6857	7	25	189/0.41	6.8	1.1	9.0	F.7	30.5	2.0	35.5	3500	440	500	0.795	55	250	440	99	2846
5013-6858	7	35	259/0.41	7.8	1.1	10.0	F.7	33.8	2.1	39.1	3500	440	500	0.565	45	250	440	120	3506
5013-6859	7	50	378/0.41	9.5	1.1	11.7	F.7	39.5	2.3	45.3	3500	440	500	0.393	45	250	440	150	3983
5013-6860	7	70	348/0.51	11.5	1.1	13.7	F.7	46.2	2.5	52.5	3500	440	500	0.277	40	250	440	189	5894
5013-6861A1	14	2.5	50 / 0.26	2.0	0.6	3.6	F.4.10	16.3	1.5	20.5	2000	440	500	8.21	100	250	440	21.5	800
5013-6861	14	4	56 / 0.31	2.5	0.6	4.1	F.4.10	18.5	1.5	22.5	2000	440	500	5.09	100	250	440	27.5	987
5013-6862	14	6	84/0.31	3.1	0.7	4.9	F.4.10	22.1	1.7	26.3	2000	440	500	3.39	100	250	440	39.5	1332
5013-6863	14	10	80/0.41	4.1	0.9	6.3	F.4.10	28.2	1.9	33.7	2000	440	500	1.95	100	250	440	50.0	2413
5013-6864	14	16	126/0.41	5.4	0.9	7.6	F.4.10	34.0	2.1	39.2	2000	440	500	1.24	100	250	440	70.0	3390
5013-6865	14	25	189/0.41	6.8	1.1	9.0	F.4.10	40.2	2.4	46.1	3500	440	500	0.795	55	250	440	82.5	4980
5013-6866	14	35	259/0.41	7.8	1.1	10.0	F.4.10	44.6	2.5	51.2	3500	440	500	0.565	45	250	440	100	5875
5013-6867A1	18	4	56 / 0.31	2.5	0.6	4.1	F.6.12	20.9	1.8	24.9	2000	440	500	5.09	100	250	440	24.8	1500
5013-6867	18	6	84 / 0.31	3.1	0.7	4.9	F.6.12	24.9	1.8	29.4	2000	440	500	3.39	100	250	440	35.6	1820
5013-6868	18	10	80 / 0.41	4.1	0.7	5.9	F.6.12	29.9	2.1	37.8	2000	440	500	1.95	100	250	440	45.0	3904
5013-6869	18	16	126/ 0.41	5.4	0.8	7.6	F.6.12	38.4	2.3	44.2	2000	440	500	1.24	100	250	440	63.0	4380
5013-6870A1	30	1.0	32/0.21	1.32	0.5	2.7	F.4.10.16	17.9	1.6	22	2000	440	500	20.0	100	250	440	9.2	840
5013-6870	30	1.5	30 / 0.26	1.55	0.5	3.0	F.4.10.16	19.3	1.6	23.6	2000	440	500	13.7	100	250	440	13.2	980
5013-6871	30	2.5	50 / 0.26	2	0.6	3.6	F.4.10.16	23.5	1.7	27.7	2000	440	500	8.21	100	250	440	17.2	1400
5013-6872	30	4	56 / 0.31	2.5	0.6	4.1	F.4.10.16	26.7	1.9	32.1	2000	440	500	5.09	100	250	440	22.0	1999
5013-6873	30	6	84 / 0.31	3.1	0.7	4.9	F.4.10.16	31.9	2.1	37.1	2000	440	500	3.39	100	250	440	31.6	2760
5013-6874	30	10	80 / 0.41	4.1	0.7	5.9	F.4.10.16	38.3	2.5	48.0	2000	440	500	1.95	100	250	440	40.0	4928

TABLES OF DIMENSIONS**PART C****TABLE C4: TWIN AND MULTI CORE CABLES**

1	2	3			4	5	6	7	8	9	10	11		12	13	14	15		16
Stock No.	No. of Cores	Conductor					No. of Cores per layer	Diameter Over taped laid up Cores (Nom)	Sheath thickness (Nom)	Overall Diameter (Nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (typical value) x Single Wire Dia Max	Diameter (Nominal)	Thickness of Insulation system (nom)	Diameter Over protective layer and Lacquer (Nom)						Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq.m m	No./mm	mm	mm	mm		mm	mm	mm	V	V	V	Ohm	Meg. Ohm	m	V	A	kg / km
5013-6828A1	2	0.75	24 / 0.21	1.1	0.5	2.5	2	5.4	1.2	8.2	2000	440	500	26.7	100	500	440	13.6	80
5013-6840A1	3	0.75	24 / 0.21	1.1	0.5	2.5	3	5.8	1.2	8.7	2000	440	500	26.7	100	500	440	11.9	100
5013-6877A1	4	0.75	24 / 0.21	1.1	0.5	2.5	F.4	6.5	1.2	9.3	2000	440	500	26.7	100	500	440	11.9	120
5013-6878A1	5	0.75	24 / 0.21	1.1	0.5	2.5	F.5	7.2	1.2	10.3	2000	440	500	26.7	100	500	440	10.2	145
5013-6879A1	7	0.75	24 / 0.21	1.1	0.5	2.5	F.7	8.9	1.2	11.7	2000	440	500	26.7	100	500	440	10.2	180
5013-6880A1	10	0.75	24 / 0.21	1.1	0.5	2.5	2.8	10.4	1.2	13.4	2000	440	500	26.7	100	500	440	8.5	220
5013-6881A1	14	0.75	24 / 0.21	1.1	0.5	2.5	F.4.10	11.5	1.3	14.6	2000	440	500	26.7	100	500	440	8.5	320
5013-6882A1	18	0.75	24 / 0.21	1.1	0.5	2.5	F.6.12	12.9	1.3	16.2	2000	440	500	26.7	100	500	440	7.6	375
5013-6883A1	24	0.75	24 / 0.21	1.1	0.5	2.5	2.8.14	15.4	1.4	18.9	2000	440	500	26.7	100	500	440	7.6	530
5013-6884A1	30	0.75	24 / 0.21	1.1	0.5	2.5	F.4.10.16	16.5	1.5	20.4	2000	440	500	26.7	100	500	440	6.8	645
5013-6885A1	36	0.75	24 / 0.21	1.1	0.5	2.5	F.6.12.18	17.9	1.5	21.8	2000	440	500	26.7	100	500	440	6.8	770
5013-6886A1	44	0.75	24 / 0.21	1.1	0.5	2.5	2.8.14.20	20.4	1.6	24.6	2000	440	500	26.7	100	500	440	6.8	925

TABLES OF DIMENSIONS**PART C****TABLE C4: TWIN AND MULTI CORE CABLES**

1	2	3			4	5	6	7	8	9	10	11		12	13	14	15		16
Stock No.	No. of Cores	Conductor			Thickness of Insulation System Diameter Over protective layer and Lacquer(Nom)		No. of Cores per layer	Diameter Over taped laid up Cores (Nom)	Sheath thickness(Nom)	Overall Diameter (Nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires(Typical value) x Single Wire Maximum Dia	Diameter(Nom)								Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq.mm	n x mm		m m	mm		mm	mm	mm	V	V	V	Ohm	Meg. Ohm	m	V	A	kg / km
5013-6828	2	1	32 / 0.21	1.3	0.5	2.7	2	5.8	1.2	8.6	2000	440	500	20	100	500	440	18.5	90
5013-6840	3	1	32 / 0.21	1.3	0.5	2.7	3	6.2	1.2	9.1	2000	440	500	20	100	500	440	16.1	115
5013-6871 (7)	4	1	32 / 0.21	1.3	0.5	2.7	F.4	6.9	1.2	9.8	2000	440	500	20	100	500	440	16.1	129
5013-6878	5	1	32 / 0.21	1.3	0.5	2.7	F.5	7.7	1.2	10.8	2000	440	500	20	100	500	440	13.8	159
5013-6879	7	1	32 / 0.21	1.3	0.5	2.7	F.7	9.6	1.2	12.4	2000	440	500	20	100	500	440	13.8	196
5013-6880	10	1	32 / 0.21	1.3	0.5	2.7	2.8	11.2	1.2	14.2	2000	440	500	20	100	500	440	11.5	239
5013-6881	14	1	32 / 0.21	1.3	0.5	2.7	F.4.10	12.3	1.3	15.5	2000	440	500	20	100	500	440	11.5	348
5013-6882	18	1	32 / 0.21	1.3	0.5	2.7	F.6.12	13.9	1.3	17.2	2000	440	500	20	100	250	440	10.3	400
5013-6883	24	1	32 / 0.21	1.3	0.5	2.7	2.8.14	16.6	1.4	20.1	2000	440	500	20	100	250	440	10.3	556
5013-6884	30	1	32 / 0.21	1.3	0.5	2.7	F.4.10.16	17.7	1.5	21.7	2000	440	500	20	100	250	440	9.2	670
5013-6885	36	1	32 / 0.21	1.3	0.5	2.7	F.6.12.18	19.3	1.5	23.2	2000	440	500	20	100	250	440	9.2	798
5013-6886	44	1	32 / 0.21	1.3	0.5	2.7	2.8.14.20	22.0	1.6	26.2	2000	440	500	20	100	250	440	9.2	960
5013-6829	2	1.5	30/0.26	1.6	0.5	2.95	2	6.3	1.2	9.1	2000	440	500	13.7	100	500	440	26.4	124
5013-6841	3	1.5	30/0.26	1.6	0.5	2.95	3	6.8	1.2	9.6	2000	440	500	13.7	100	500	440	23.1	138
5013-6889A1	4	1.5	30/0.26	1.6	0.5	2.95	F.4	7.5	1.2	10.4	2000	440	500	13.7	100	250	440	23.1	186
5013-6890A1	5	1.5	30/0.26	1.6	0.5	2.95	F.5	8.4	1.2	11.5	2000	440	500	13.7	100	250	440	19.8	225
5013-6891A1	7	1.5	30/0.26	1.6	0.5	2.95	F.7	10.4	1.3	13.5	2000	440	500	13.7	100	250	440	19.8	319
5013-6892A1	10	1.5	30/0.26	1.6	0.5	2.95	2.8	12.2	1.4	15.6	2000	440	500	13.7	100	250	440	16.5	398

TABLES OF DIMENSIONS**PART C****TABLE C4: TWIN AND MULTI CORE CABLES**

1	2	3			4	5	6	7	8	9	10	11		12	13	14	15		16
Stock No.	No. of Cores	Conductor					No. of Cores per layer	Diameter Over taped laid up Cores (nom)	Sheath thickness (nom)	Overall Diameter (Nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (typical value) x Single Wire Maximum Dia	Diameter(nom)	Thickness of Insulation System(nom)	Diameter Over protective layer and Lacquer (nom)						Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
5013-6893A1	14	1.5	30 / 0.26	1.6	0.5	2.95	F.4.10	13.4	1.4	16.8	2000	440	500	13.7	100	250	440	16.5	509
5013-6894A1	18	1.5	30 / 0.26	1.6	0.5	2.95	F.6.12	15.2	1.5	18.9	2000	440	500	13.7	100	250	440	14.8	641
5013-6895A1	24	1.5	30 / 0.26	1.6	0.5	2.95	2.8.14	18.1	1.7	22.2	2000	440	500	13.7	100	250	440	14.8	859
5013-6896A1	30	1.5	30 / 0.26	1.6	0.5	2.95	F.4.10.16	19.3	1.7	23.7	2000	440	500	13.7	100	250	440	13.2	1027
5013-6897A1	36	1.5	30 / 0.26	1.6	0.5	2.95	F.6.12.18	21.1	1.8	25.6	2000	440	500	13.7	100	250	440	13.2	1222
5013-6898A1	44	1.5	30 / 0.26	1.6	0.5	2.95	2.8.14.20	24.0	2.0	29.0	2000	440	500	13.7	100	250	440	13.2	1487
5013-6830	2	2.5	50 / 0.26	2	0.6	3.6	2	7.6	1.2	10.5	2000	440	500	8.21	100	500	440	34.4	168
5013-6842	3	2.5	50 / 0.26	2	0.6	3.6	3	8.2	1.2	11.1	2000	440	500	8.21	100	500	440	30.1	205
5013-6889	4	2.5	50 / 0.26	2	0.6	3.6	F.4	9.1	1.2	12.0	2000	440	500	8.21	100	250	440	30.1	255
5013-6890	5	2.5	50 / 0.26	2	0.6	3.6	F.5	10.1	1.2	13.0	2000	440	500	8.21	100	250	440	25.8	323
5013-6891	7	2.5	50 / 0.26	2	0.6	3.6	F.7	12.6	1.3	15.7	2000	440	500	8.21	100	250	440	25.8	419
5013-6892	10	2.5	50 / 0.26	2	0.6	3.6	2.8	14.8	1.4	18.2	2000	440	500	8.21	100	250	440	21.5	544

TABLES OF DIMENSIONS

PART C

TABLE C4: TWIN AND MULTI CORE CABLES

1	2	3			4	5	6	7	8	9	10	11		12	13	14	15		16
Stock No.	No. of Cores	Conductor			Thickness of Insulation System (Nom)	Diameter Over protective covering and Lacquer (nom)	No. of Cores per layer	Diameter Over taped laid up Cores ((Nom)	Sheath thickness (Nom)	Overall Diameter (nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires(typical) x Single Wire Maximum Dia	Diameter (Nom)								Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq. mm	NO. x mm	mm	mm	mm	No.	mm	mm	mm	V	V	V	Ohm	Meg. Ohm	m	V	A	kg / km
5013-6893	14	2.5	50 / 0.26	2	0.6	3.6	F.4.10	16.3	1.4	19.7	2000	440	500	8.21	100	250	440	21.5	697
5013-6894	18	2.5	50 / 0.26	2	0.6	3.6	F.6.12	18.4	1.5	22.1	2000	440	500	8.21	100	250	440	19.3	910
5013-6895	24	2.5	50 / 0.26	2	0.6	3.6	2.8.14	22.0	1.7	26.1	2000	440	500	8.21	100	250	440	19.3	1171
5013-6871	30	2.5	50 / 0.26	2	0.6	3.6	F.4.10.16	23.5	1.7	27.6	2000	440	500	8.21	100	250	440	17.2	1400
5013-6897	36	2.5	50 / 0.26	2	0.6	3.6	F.6.12.18	25.6	1.8	30.2	2000	440	500	8.21	100	250	440	17.2	1600
5013-6898	44	2.5	50 / 0.26	2	0.6	3.6	2.8.14.20	29.2	2.0	34.2	2000	440	500	8.21	100	250	440	17.2	2042
5013-6831	2	4.0	56 / 0.31	2.6	0.6	4.15	2	8.7	1.2	11.6	2000	440	500	5.09	100	500	440	44.0	204
5013-6843	3	4.0	56 / 0.31	2.6	0.6	4.15	3	9.4	1.2	12.3	2000	440	500	5.09	100	500	440	38.5	249
5013-6901	4	4.0	56 / 0.31	2.6	0.6	4.15	F.4	10.4	1.2	13.3	2000	440	500	5.09	100	250	440	38.5	308
5013-6902	5	4.0	56 / 0.31	2.6	0.6	4.15	F.5	11.6	1.2	14.5	2000	440	500	5.09	100	250	440	33.0	371
5013-6903	7	4.0	56 / 0.31	2.6	0.6	4.15	F.7	14.5	1.4	17.9	2000	440	500	5.09	100	250	440	33.0	523
5013-6904	10	4.0	56 / 0.31	2.6	0.6	4.15	2.8	17.0	1.5	20.7	2000	440	500	5.09	100	250	440	27.5	719
5013-6861	14	4.0	56 / 0.31	2.6	0.6	4.15	F.4.10	18.7	1.5	22.4	2000	440	500	5.09	100	250	440	27.5	923
5013-6906	18	4.0	56 / 0.31	2.6	0.6	4.15	F.6.12	21.2	1.6	25.1	2000	440	500	5.09	100	250	440	24.7	1251
5013-6907	24	4.0	56 / 0.31	2.6	0.6	4.15	2.8.14	25.3	1.8	29.6	2000	440	500	5.09	100	250	440	24.7	1546
5013-6872	30	4.0	56 / 0.31	2.6	0.6	4.15	F.4.10.16	27.0	1.9	31.8	2000	440	500	5.09	100	250	440	22.0	1865
5013-6909	36	4.0	56 / 0.31	2.6	0.6	4.15	F.6.12.18	29.5	2	34.5	2000	440	500	5.09	100	250	440	22.0	2313

TABLES OF DIMENSIONS
PART C
TABLE C5: MULTIPAIRED CABLES COLLECTIVELY SCREENED

1	2	3			4	5	6	7	8	9	10	11	12		13	14	15	16		17
Stock No.	No. of Pairs	Conductor			Thickness of Insulation System (Nom)	Diameter Over protective layer and Lacquer(Nom)	Diameter Over taped pairs(Nom)	No.of Pairs per layer	Screen Wire Diameter(Nom)	Sheath thickness(Nom)	Overall Diameter (Nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (Typical) x Single Wire Maximum Dia.	Diameter(Nom)									Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq.m m	No x mm	mm	mm	mm	mm		mm	mm	mm	V	V	V	Ohm	Meg. Ohm	m	V	A	kg / km
5013-6910A1	1	0.75	24 / 0.21	1.1	2.52	2.72	5.4	1	0.15	1.2	8.8	2000	440	250	26.7	100	250	440	13.6	100
5013-6911A1	2	0.75	24 / 0.21	1.1	2.52	2.72	5.4	2	0.2	1.2	13.6	2000	440	250	26.7	100	250	440	11.9	236
5013-6912A1	3	0.75	24 / 0.21	1.1	2.52	2.72	5.4	3	0.2	1.2	14.4	2000	440	250	26.7	100	250	440	10.2	241
5013-6913A1	4	0.75	24 / 0.21	1.1	2.52	2.72	5.4	4	0.2	1.3	15.0	2000	440	250	26.7	100	250	440	10.2	352
5013-6914A1	7	0.75	24 / 0.21	1.1	2.52	2.72	5.4	1.6	0.2	1.4	17.7	2000	440	250	26.7	100	250	440	8.5	489
5013-6915A1	11	0.75	24 / 0.21	1.1	2.52	2.72	5.4	3.8	0.3	1.6	23.7	2000	440	250	26.7	100	250	440	7.6	663
5013-6916A1	16	0.75	24 / 0.21	1.1	2.52	2.72	5.4	5.11	0.3	1.7	26.0	2000	440	250	26.7	100	250	440	6.8	829
5013-6917A1	19	0.75	24 / 0.21	1.1	2.52	2.72	5.4	1.6.12	0.3	1.7	27.4	2000	440	250	26.7	100	250	440	6.8	962
5013-6910	1	1.0	32 / 0.21	1.3	2.72	2.92	5.8	1	0.15	1.2	9.2	2000	440	250	20	100	250	440	18.5	104
5013-6911	2	1.0	32 / 0.21	1.3	2.72	2.92	5.8	2	0.2	1.2	14.3	2000	440	250	20	100	250	440	16.1	245
5013-6912	3	1.0	32 / 0.21	1.3	2.72	2.92	5.8	3	0.2	1.2	15.2	2000	440	250	20	100	250	440	13.8	254
5013-6913	4	1.0	32 / 0.21	1.3	2.72	2.92	5.8	4	0.2	1.3	15.8	2000	440	250	20	100	250	440	13.8	370
5013-6914	7	1.0	32 / 0.21	1.3	2.72	2.92	5.8	1.6	0.2	1.4	18.7	2000	440	250	20	100	250	440	11.5	520
5013-6915	11	1.0	32 / 0.21	1.3	2.72	2.92	5.8	3.8	0.3	1.6	25.0	2000	440	250	20	100	250	440	10.3	712
5013-6916	16	1.0	32 / 0.21	1.3	2.72	2.92	5.8	5.11	0.3	1.7	27.6	2000	440	250	20	100	250	440	9.2	900
5013-6917	19	1.0	32 / 0.21	1.3	2.72	2.92	5.8	1.6.12	0.3	1.7	29.0	2000	440	250	20	100	250	440	9.2	1046

TABLES OF DIMENSIONS

PART C

TABLE C6: MULTICORE CABLES INDIVIDUALLY SCREENED CORES

1	2	3			4	5	6	7	8	9	10	11	12	13		14	15	16	17		18
Stock No.	No. of Cores	Conductor			Thickness of Insulation System(Nom)	Diameter Over protective layer and Lacquer(Nom)	Screen Wire Diameter(Nom)	Diameter Over Insulated Screen(Nom)	No.of Cores per layer	Diameter Over taped Cores(Nom)	Sheath radial thickness(Nom)	Overall Diameter(Nom)	Test voltage	Gas Flame Test		Max. Conductor Resistance per km at 20°C	Min. Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires(Typical) x Single Wire Maximum Dia	Diameter(Nom)										Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq.m m	no x mm	mm	mm	mm	mm	mm		mm	mm	mm	V	V	V	Ohm	Meg. Ohm	m	V	A	kg / km
5013-6918A1	3	0.75	24/0.21	1.1	0.5	2.5	0.15	3.4	3	7.7	1.2	10.7	2000	440	250	26.7	100	250	440	11.9	203
5013-6919A1	5	0.75	24/0.21	1.1	0.5	2.5	0.15	3.4	F.5	9.5	1.2	12.5	2000	440	250	26.7	100	250	440	10.2	254
5013-6920A1	7	0.75	24/0.21	1.1	0.5	2.5	0.15	3.4	F.7	11.8	1.3	14.9	2000	440	250	26.7	100	250	440	10.2	363
5013-6921A1	14	0.75	24/0.21	1.1	0.5	2.5	0.15	3.4	F. 4.10	15.3	1.4	18.7	2000	440	250	26.7	100	250	440	8.5	569
5013-6922A1	18	0.75	24/0.21	1.1	0.5	2.5	0.15	3.4	F. 6.12	17.3	1.5	21.2	2000	440	250	26.7	100	250	440	7.6	710
5013-6918	3	1.0	32/0.21	1.32	0.5	2.72	0.15	3.6	3	8.1	1.2	11.1	2000	440	250	20	100	250	440	16.1	210
5013-6919	5	1.0	32/0.21	1.32	0.5	2.72	0.15	3.6	F.5	10.0	1.2	13.0	2000	440	250	20	100	250	440	13.8	265
5013-6920	7	1.0	32/0.21	1.32	0.5	2.72	0.15	3.6	F.7	12.4	1.3	15.5	2000	440	250	20	100	250	440	13.8	379
5013-6921	14	1.0	32/0.21	1.32	0.5	2.72	0.15	3.6	F. 4.10	16.2	1.4	19.6	2000	440	250	20	100	250	440	11.5	600
5013-6922	18	1.0	32/0.21	1.32	0.5	2.72	0.15	3.6	F. 6.12	18.3	1.5	22.2	2000	440	250	20	100	250	440	10.3	750

2. Min. Insulation Resistance between individual screens per km at 20°C at 500V - 2 megohm

TABLES OF DIMENSIONS

PART C

TABLE C7 : MULTIPAIRED, CABLES, PAIRS INDIVIDUALLY SCREENED

1	2	3			4	5	6	7	8	9	10	11	12	13	14		15	16	17	18		19
Stock No.	No. of Pairs	Conductor			Thickness of Insulation System (Nom)	Diameter Over protective layer and Lacquer(Nom)	Diameter over taped twisted Pair(Nom)	Screen Wire Diameter(Nom)	Diameter Over Insulated Screen(Nom)	No. of Pairs per Layer	Diameter Over Laid up and taped pairs(Nom)	Sheath thickness (Nom)	Overall Diameter(Nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (Typical value) x Single Wire Maximum Dia	Diameter(Nom)											Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq. mm	no x mm	mm	mm	mm	mm	mm	mm		mm	mm	mm	V	V	V	Ohm	Meg. Ohm	m	V	A	kg / km
5013-6910A1	1	0.75	24/0.21	1.1	0.5	2.5	5.4	0.15	6	1	-	1.2	8.7	2000	250	250	26.7	100	250	440	13.6	94
5013-6924A1	3	0.75	24/0.21	1.1	0.5	2.5	5.4	0.15	6	3	11.2	1.3	14.0	2000	250	250	26.7	100	250	440	10.2	230
5013-6925A1	4	0.75	24/0.21	1.1	0.5	2.5	5.4	0.15	6	4	12.5	1.3	15.3	2000	250	250	26.7	100	250	440	10.2	300
5013-6926A1	7	0.75	24/0.21	1.1	0.5	2.5	5.4	0.15	6	1.6	15.6	1.4	18.5	2000	250	250	26.7	100	250	440	8.5	460
5013-6910	1	1.0	32/0.21	1.32	0.5	2.72	5.9	0.15	6.5	1	-	1.2	9.2	2000	250	250	20	100	250	440	18.5	104
5013-6924	3	1.0	32/0.21	1.32	0.5	2.72	5.9	0.15	6.5	3	11.7	1.3	14.9	2000	250	250	20	100	250	440	13.8	251
5013-6925	4	1.0	32/0.21	1.32	0.5	2.72	5.9	0.15	6.5	4	12.9	1.3	16.3	2000	250	250	20	100	250	440	13.8	323
5013-6926	7	1.0	32/0.21	1.32	0.5	2.72	5.9	0.15	6.5	1.6	16.2	1.4	19.7	2000	250	250	20	100	250	440	11.5	500

Min. Insulation Resistance between individual screens per km at 20°C at 500V - 2 megohm

TABLES OF DIMENSIONS**PART C****TABLE C8: MULTICORE CABLES, COLLECTIVELY SCREENED**

1	2		3			4	5	6	7	8	9	10	11		12	13	14	15		16
Stock No.	No. of Cores		Conductor			Thickness of Insulation System(Nom)	Diameter Over protective layer and Lacquer(Nom)	No. of Cores per layer	Screen Wire Diameter(Nom)	Sheath thickness(Nom)	Overall Diameter(Nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
			Nominal Area	No. of Wires (Typical) x Single Wire Max. Dia.	Diameter(Nom)								Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
			Sq mm	No x mm	mm	mm	mm		mm	mm	mm	V	V	V	Ohm	Mega.ohm	m	V	A	kg / km
5013-6910A1	2		0.75	24/0.21	1.12	0.5	2.52	2	0.15	1.2	8.6	2000	440	440	26.7	100	250	440	15	104
5013-6928A1	8		0.75	24/0.21	1.12	0.5	2.52	F.8	0.2	1.2	13.7	2000	440	440	26.7	100	250	440	11	292
5013-6929A1	12		0.75	24/0.21	1.12	0.5	2.52	3.9	0.2	1.3	15.2	2000	440	440	26.7	100	250	440	9	349
5013-6930A1	14		0.75	24/0.21	1.12	0.5	2.52	F.4.10	0.2	1.3	15.8	2000	440	440	26.7	100	250	440	9	370
5013-6910	2		1.0	32/0.21	1.32	0.5	2.72	2	0.15	1.2	9.2	2000	440	440	20	100	250	440	18	120
5013-6928	8		1.0	32/0.21	1.32	0.5	2.72	F.8	0.2	1.2	14.4	2000	440	440	20	100	250	440	14	310
5013-6929	12		1.0	32/0.21	1.32	0.5	2.72	3.9	0.2	1.3	16	2000	440	440	20	100	250	440	12	390
5013-6930	14		1.0	32/0.21	1.32	0.5	2.72	F.4.10	0.2	1.3	16.7	2000	440	440	20	100	250	440	12	432
5013-6931A1	4		1.5	30/0.26	1.6	0.5	3	F.4	0.2	1.2	12	2000	440	440	13.7	100	250	440	23	250
5013-6931	4		2.5	50/0.26	2	0.5	3.4	F.4	0.2	1.2	13	2000	440	440	8.21	100	250	440	30	278
5013-6932A1	2		2.5	50/0.26	2.1	0.5	3.5	2	0.2	1.2	11.4	2000	440	440	8.21	100	250	440	33	215
5013-6932	2		4.0	56/0.31	2.5	0.6	4.1	2	0.2	1.2	12.6	2000	440	440	5.09	100	250	440	44	249
5013-6933A1	6 8	}14	0.75	24/0.21	1.12	0.5	2.52								26.7				10	
			1.5	30/0.26	1.6	0.5	3	F.6.8	0.3	1.5	19.5	2000	440	440	13.7	100	250	440	20	550
5013-6933	6 8	}14	1	32/0.21	1.32	0.5	2.72								20				12	
			2.5	50/0.26	2	0.5	3.4	F.6.8	0.3	1.5	21.1	2000	440	440	8.21	100	250	440	22	653
5013-5869A1	2		6	84/0.31	3.1	0.7	5.1	2	0.2	1.5	15.2	2000	440	440	3.39	100	250	440	64	400
5013-5869	2		10	80/0.41	4.1	0.7	6.1	2	0.2	1.5	17.7	2000	440	440	1.95	100	250	440	80	500

Notes : The above current ratings are for each core when all cores are equally loaded.

TABLES OF DIMENSIONS**PART C****TABLE C9: MINIATURE MULTICORE CABLES, UNSCREENED (OR) INDIVIDUALLY (OR) COLLECTIVELY SCREENED**

1	2	3			4	5	6	7	8	9	10	11	12	13	14	15		16	17	18	19		20
Stock No.	No. of Cores	Conductor			Thickness of Insulation System(Nom)	Diameter Over protective layer and Lacquer(Nom)	Screen Wire Diameter(Nom)	Diameter Over Insulated Screen(nom)	No. of Cores per layer	Diameter Over taped laid up cores (Nom)	Screen Wire Diameter(Nom)	Diameter Over screen laid up cores(Nom)	Sheath thickness (Nom)	Overall Diameter (Nom)	Voltage test	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Min. insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (typical) x Single Wire Maximum Dia	Diameter(Nom)												Test Voltage	Min. Breakdown Voltage				Voltage	Current	
		Sq. mm	No x mm	mm	mm	mm	mm	mm		mm	mm	mm	mm	mm	V	V	V	Ohm	Meg. Ohm	m	V	A	Kg/ KM
5013-6970	2	0.35	19 / 0.15	0.75	0.35	1.65	x	x	2	3.75	x	x	1.0	6.2	2000	250	250	60	100	100	440	6.4	50
5013-6971	2	0.35	19 / 0.15	0.75	0.35	1.65	x	x	2	3.75	0.15	4.5	1.0	7.0	2000	250	250	60	100	100	440	6.4	74
5013-6972	2	0.35	19 / 0.15	0.75	0.35	1.65	0.1	2.45	2	4.95	x	x	1.0	7.4	2000	250	250	60	100	100	440	6.4	81
5013-6973	3	0.35	19 / 0.15	0.75	0.35	1.65	x	x	3	4.05	x	x	1.0	6.5	2000	250	250	60	100	100	440	5.6	58
5013-6974	3	0.35	19 / 0.15	0.75	0.35	1.65	x	x	3	4.05	0.15	4.8	1.0	7.3	2000	250	250	60	100	100	440	5.6	81
5013-6975	3	0.35	19 / 0.15	0.75	0.35	1.65	0.1	2.45	3	5.35	x	x	1.0	7.8	2000	250	250	60	100	100	440	5.6	93
5013-6976	4	0.35	19 / 0.15	0.75	0.35	1.65	x	x	F.4	4.55	x	x	1.0	7.0	2000	250	250	60	100	100	440	5.6	66
5013-6977	4	0.35	19 / 0.15	0.75	0.35	1.65	x	x	F.4	4.55	0.15	5.3	1.0	7.8	2000	250	250	60	100	100	440	5.6	92
5013-6978	4	0.35	19 / 0.15	0.75	0.35	1.65	0.1	2.45	F.4	5.95	x	x	1.0	8.4	2000	250	250	60	100	100	440	5.6	110
5013-6979	6	0.35	19 / 0.15	0.75	0.35	1.65	x	x	F.6	5.6	x	x	1.0	8.1	2000	250	250	60	100	100	440	4.8	87
5013-6980	6	0.35	19 / 0.15	0.75	0.35	1.65	x	x	F.6	5.6	0.15	6.35	1.1	9.0	2000	250	250	60	100	100	440	4.8	124
5013-6981	6	0.35	19 / 0.15	0.75	0.35	1.65	0.1	2.45	F.6	7.4	x	x	1.1	10.1	2000	250	250	60	100	100	440	4.8	154
5013-6982	12	0.35	19 / 0.15	0.75	0.35	1.65	x	x	3.9	7.75	x	x	1.1	10.4	2000	250	250	60	100	100	440	4.0	148
5013-6983	12	0.35	19 / 0.15	0.75	0.35	1.65	x	x	3.9	7.75	0.15	8.5	1.1	11.3	2000	250	250	60	100	100	440	4.0	191
5013-6984	12	0.35	19 / 0.15	0.75	0.35	1.65	0.1	2.45	3.9	10.25	x	x	1.2	13.2	2000	250	250	60	100	100	440	4.0	270
5013-6985	18	0.35	19 / 0.15	0.75	0.35	1.65	x	x	F.6.12	9.3	x	x	1.2	12.3	2000	250	250	60	100	100	440	3.6	204
5013-6986	18	0.35	19 / 0.15	0.75	0.35	1.65	x	x	F.6.12	9.3	0.2	10.3	1.2	13.3	2000	250	250	60	100	100	440	3.6	270
5013-6987	18	0.35	19 / 0.15	0.75	0.35	1.65	0.1	2.45	F.6.12	12.3	x	x	1.3	15.5	2000	250	250	60	100	100	440	3.6	379

TABLES OF DIMENSIONS

PART C

TABLE C9: MINIATURE MULTICORE CABLES, UNSCREENED (OR) INDIVIDUALLY (OR) COLLECTIVELY SCREENED

1	2	3			4	5	6	7	8	9	10	11	12	13	14	15		16	17	18	19		20
Stock No.	No. of Cores	Conductor			Insulation thickness(Nom)	Diameter Over protective layer and Lacquer(Nom)	Screen Wire Diameter(Nom)	Diameter Over Insulated Screen(Nom)	No.of Cores per layer (nom)	Diameter Over taped laid up cores (nom)	Screen Wire Diameter(nom)	Diameter Over screen laid up cores(nom)	Sheath thickness (nom)	Overall Diameter(nom)	Voltage test	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Min. insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (Typical) x Single Wire Maximum Dia	Diameter(Nom)												Test Voltage	Min. Breakdown Voltage				Voltage	Current	
		Sq. mm	No x mm	mm	mm	mm	mm	mm		mm	mm	mm	mm	mm	V	V	V	ohm	Meg. Ohm	m	V	A	Kg/Km
5013-6988	25	0.35	19 / 0.15	0.75	0.35	1.65	x	x	2.8.15	11.15	x	x	1.2	14.1	2000	250	250	60	100	100	440	3.6	275
5013-6989	25	0.35	19 / 0.15	0.75	0.35	1.65	x	x	2.8.15	11.15	0.2	12.15	1.3	15.3	2000	250	250	60	100	100	440	3.6	360
5013-6990	25	0.35	19 / 0.15	0.75	0.35	1.65	0.1	2.45	2.8.15	14.75	x	x	1.4	18.2	2000	250	250	60	100	100	440	3.6	524
5013-6991	36	0.35	19 / 0.15	0.75	0.35	1.65	x	x	F.6. 12.18	13	0.2	14	1.4	17.5	2000	250	250	60	100	100	440	3.2	465
5013-6992	60	0.35	19 / 0.15	0.75	0.35	1.65	x	x	F.6.12.18.24	16.7	x	x	1.5	20.4	2000	250	250	60	100	100	440	2.4	590
5013-6993	60	0.35	19 / 0.15	0.75	0.35	1.65	x	x	F.6.12.18.24	16.7	0.2	17.7	1.5	21.6	2000	250	250	60	100	100	440	2.4	702

TABLES OF DIMENSIONS**PART C****TABLE C10: MINIATURE PAIR AND TRIPLE CABLES, COLLECTIVELY SCREENED**

1	2	3			4	5	6	7	8	9	10	11	12	13	14		15	16	17	18		19
Stock No.	No. of Pairs (or) Triples	Conductor			Insulation thickness (nom)	Dia Over protective covering and Lacquer(nom)	Diameter Over taped pair/ Triple(Nom)	No. of Pairs / Triples per layer	Diameter Over taped laid up Pairs / Triples (nom)	Screen Wire Diameter(nom)	Diameter Over screen Pairs (or) Triples(nom)	Sheath thickness(nom)	Overall Diameter(nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of cable
		Nominal Area	No. of Wires (typical) x Single Wire Maximum Dia	Diameter(nom)											Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq.mm	No x mm	mm	mm	mm	mm		mm	mm	mm	mm	mm	V	V	V	ohm	Meg. Ohm	m	V	A	Kg/Km
5013-6971	1P	0.35	19 / 0.15	0.75	0.35	1.85	3.75	1	3.75	0.15	4.5	1.0	7.0	2000	250	250	60	100	100	440	6.4	68
5013-6995	2P	0.35	19 / 0.15	0.75	0.35	1.85	3.75	2	6.2	0.15	6.95	1.1	9.6	2000	250	250	60	100	100	440	5.6	114
5013-6996	3P	0.35	19 / 0.15	0.75	0.35	1.85	3.75	3	6.75	0.15	7.5	1.1	10.2	2000	250	250	60	100	100	440	4.8	133
5013-6997	5P	0.35	19 / 0.15	0.75	0.35	1.85	3.75	F.5	8.4	0.15	9.15	1.2	12.1	2000	250	250	60	100	100	440	4.0	187
5013-6998	11P	0.35	19 / 0.15	0.75	0.35	1.85	3.75	3.8	12.85	0.2	13.85	1.4	17.3	2000	250	250	60	100	100	440	3.6	365
5013-6999	25P	0.35	19 / 0.15	0.75	0.35	1.85	3.75	2.8.15	17.9	0.2	18.9	1.6	23.0	2000	250	250	60	100	100	440	2.4	664
5013-6974	1T	0.35	19 / 0.15	0.75	0.35	1.85	4.05	1	4.05	0.15	4.8	1.0	7.3	2000	250	250	60	100	100	440	5.6	78
5013-7001	2T	0.35	19 / 0.15	0.75	0.35	1.85	4.1	2	8.2	0.15	8.95	1.2	11.9	2000	250	250	60	100	100	440	4.8	149
5013-7002	3T	0.35	19 / 0.15	0.75	0.35	1.85	4.1	F.3	8.45	0.15	9.2	1.2	12.2	2000	250	250	60	100	100	440	4.0	188
5013-7003	4T	0.35	19 / 0.15	0.75	0.35	1.85	4.1	F.4	9.45	0.2	10.45	1.2	13.4	2000	250	250	60	100	100	440	4.0	237
5013-7004	7T	0.35	19 / 0.15	0.75	0.35	1.85	4.1	1.6	11.5	0.2	12.5	1.3	15.8	2000	250	250	60	100	100	440	3.6	320

TABLES OF DIMENSIONS

PART C

TABLE C11: MINIATURE MULTICORE CABLES, INDIVIDUALLY AND COLLECTIVELY SCREENED CORES

1	2	3			4	5	6	7	8	9	10	11	12	13	14	15		16	17	18	19		20
Stock No.	No. of Cores	Conductor			Insulation thickness(nom)	Diameter Over protective layer and Lacquer(nom)	Screen Wire Diameter(nom)	Diameter Over Insulated screen (nom)	No .of Cores per layer	Diameter Over taped laid up cores(nom)	Screen Wire Diameter(nom)	Diameter Over screened laid up cores (nom)	Sheath thickness(nom)	Overall Diameter (nom)	Test voltage	Gas Flame Test		Max. Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of cable
		Nominal Area	No. of Wires(typical) x Single Wire Maximum Dia	Diameter(nom)												Test Voltage	Min. Breakdown Voltage				Voltage	Current	
		Sq. mm	No x mm	mm	mm	mm	mm	mm		mm	mm	mm	mm	mm	V	V	V	Ohm	Meg. Ohm	m	V	A	Kg/ Km
5013-1430	6	0.35	19 / 0.15	0.75	0.35	1.85	0.1	2.45	F.6	7.45	0.15	6.2	1.1	11.0	2000	250	250	60	100	250	440	4.8	192
5013-1431	12	0.35	19 / 0.15	0.75	0.35	1.85	0.1	2.45	3.9	10.3	0.2	11.3	1.3	14.5	2000	250	250	60	100	100	440	4.0	335
5013-1432	18	0.35	19 / 0.15	0.75	0.35	1.85	0.1	2.45	F.6.12	12.35	0.2	13.35	1.3	16.7	2000	250	250	60	100	100	440	3.6	445
5013-1433	25	0.35	19 / 0.15	0.75	0.35	1.85	0.1	2.45	2.8.15	14.8	0.2	15.8	1.5	19.5	2000	250	250	60	100	100	440	3.2	599

Min. Insulation Resistance between individual screens per km at 20°C at 500V not less than 2 megohm.

TABLES OF DIMENSIONS
TABLE C12: TWIN AND THREE CORE CABLES, OVERALL SCREENED

1	2	3			4	5	6	7	8	9	10	11	12		13	14	15	16		17
Stock No.	No. of Cores	Conductor			Thickness of Insulation System(Nom)	Dia over core (Nom)	Diameter Over protective layer and Lacquer(Nom)	Sheath thickness(Nom)	Diameter over sheath (Nom)	Screen Wire Diameter(Nom)	Overall Diameter(Nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (Typical) x Single Wire Max. Dia.	Diameter(Nom)									Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq mm	mm	mm	mm	mm	mm	Mm	mm	mm	mm	V	V	V	Ohm	Mega.ohm	m	V	A	kg / km
5013-6828A1S	2	0.75	24/0.21	1.1	0.5	2.1	2.5	1.2	7.6	0.3	9.7	2000	440	500	26.7	100	250	440	13.6	143
5013-6828S	2	1	32/0.21	1.3	0.5	2.3	2.7	1.2	8.0	0.3	10.1	2000	440	500	20.0	100	250	440	18.5	152
5013-6829S	2	1.5	30/0.26	1.55	0.5	2.55	2.95	1.2	8.5	0.3	10.5	2000	440	500	13.7	100	250	440	26.5	205
5013-6830S	2	2.5	50/0.26	2	0.6	3.2	3.6	1.2	9.8	0.3	11.9	2000	440	500	8.21	100	250	440	34.5	260
5013-6831S	2	4	56/0.31	2.5	0.6	3.7	4.1	1.2	10.8	0.3	13.0	2000	440	500	5.09	100	250	440	44.0	318
5013-6832S	2	6	84/0.31	3.1	0.7	4.5	4.9	1.2	12.4	0.3	14.6	2000	440	500	3.39	100	250	440	63.2	411
5013-6833S	2	10	80/0.41	4.1	0.7	5.5	5.9	1.3	14.6	0.3	17.7	2000	440	500	1.95	100	250	440	80.0	622
5013-6834S	2	16	126/0.41	5.4	0.8	7.0	7.6	1.4	18.2	0.3	20.5	2000	440	500	1.24	100	250	440	112	829
5013-6835S	2	25	189/0.41	6.8	0.8	8.4	9	1.5	21.2	0.3	23.7	3500	440	500	0.795	55	250	440	132	1038
5013-6836S	2	35	259/0.41	7.8	0.8	9.4	10	1.6	23.4	0.3	25.9	3500	440	500	0.565	45	250	440	160	1242
5013-6837S	2	50	378/0.41	9.5	0.8	11.1	11.7	1.7	27.0	0.3	29.5	3500	440	500	0.393	45	250	440	200	1578
5013-6838S	2	70	348/0.51	11.5	0.8	13.1	13.7	1.8	31.2	0.4	34.2	3500	440	500	0.277	40	100	440	252	2777
5013-6839S	2	95	456/0.51	12.9	1.0	14.9	15.7	2.0	35.6	0.4	38.8	3500	440	500	0.21	40	100	440	304	2913

Note: If the Copper wire braiding is required below the sheath, then for sizes 35 Sq.mm and above the diameter will be increased by 1 mm.

TABLES OF DIMENSIONS**PART C****TABLE C12: TWIN AND THREE CORE CABLES, OVERALL SCREENED**

1	2	3			4	5	6	7	8	9	10	11	12		13	14	15	16		17
Stock No.	No. of Cores	Conductor			Thickness of Insulation System(Nom)	Dia over core (Nom)	Diameter Over protective layer and Lacquer(Nom)	Sheath thickness(Nom)	Diameter over sheath (Nom)	Screen Wire Diameter(Nom)	Overall Diameter(Nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (Typical) x Single Wire Max. Dia.	Diameter(Nom)									Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	V	V	V	Ohm	Mega.ohm	m	V	A	kg / km
5013-6840A1S	3	0.75	24/0.21	1.1	0.5	2.1	2.5	1.2	8.0	0.3	10.0	2000	440	500	26.7	100	250	440	11.9	160
5013-6840S	3	1	32/0.21	1.3	0.5	2.3	2.7	1.2	8.4	0.3	10.6	2000	440	500	20.0	100	250	440	16.1	174
5013-6841S	3	1.5	30/0.26	1.6	0.5	2.55	2.95	1.2	9.0	0.3	11.0	2000	440	500	13.7	100	250	440	23.1	223
5013-6842S	3	2.5	50/0.26	2.0	0.6	3.2	3.6	1.2	10.4	0.3	12.6	2000	440	500	8.21	100	250	440	30.1	302
5013-6843S	3	4	56/0.31	2.5	0.6	3.7	4.1	1.2	11.5	0.3	13.6	2000	440	500	5.09	100	250	440	38.5	367
5013-6844S	3	6	84/0.31	3.1	0.7	4.5	4.9	1.2	13.2	0.3	15.4	2000	440	500	3.39	100	250	440	55.3	483
5013-6846S	3	10	80/0.41	4.1	0.7	5.5	5.9	1.4	15.7	0.3	18.9	2000	440	500	1.95	100	250	440	70.0	775
5013-6847S	3	16	126/0.41	5.4	0.8	7	7.6	1.4	19.4	0.3	21.6	2000	440	500	1.24	100	250	440	98.0	995
5013-6848S	3	25	189/0.41	6.8	0.8	8.9	9.5	1.6	23.9	0.3	25.3	3500	440	500	0.795	55	250	440	115	1400
5013-6849S	3	35	259/0.41	7.8	0.8	10	10.6	1.6	26.3	0.3	27.4	3500	440	500	0.565	45	250	440	140	1888
5013-6850S	3	50	378/0.41	9.5	0.8	11.7	12.3	1.8	30.4	0.3	31.5	3500	440	500	0.393	45	250	440	175	2096
5013-6851S	3	70	348/0.51	11.5	0.8	13.9	14.5	1.9	35.3	0.4	36.7	3500	440	500	0.277	40	100	440	220	2868
5013-6852S	3	95	456/0.51	12.9	1.0	15.7	16.5	2.1	40.0	0.4	41.5	3500	440	500	0.21	40	100	440	266	4301
5013-6853S	3	120	570/0.51	14.8	1.0	17.6	18.4	2.3	44.5	0.4	46.2	3500	440	500	0.164	40	100	440	294	5109
5013-6854S	3	150	741/0.51	16.8	1.0	19.6	20.4	2.4	49.1	0.4	50.7	3500	440	500	0.132	35	100	440	336	6067
5013-6855S	3	185	893/0.51	18.5	1.0	21.3	22.1	2.6	53.1	0.4	54.8	3500	440	500	0.108	30	100	440	392	7035
5013-6856S	3	240	1184/0.51	21.0	1.0	23.8	24.6	2.8	58.9	0.4	60.7	3500	440	500	0.082	25	100	440	462	9492

Note: If the Copper wire braiding is required below the sheath, then for sizes 35 Sq.mm and above the diameter will be increased by 1 mm

TABLES OF DIMENSIONS**PART C****TABLE C13: MULTI CORE CABLES, OVERALL SCREENED**

1	2	3			4	5	6	7	8	9	10	11	12		13	14	15	16		17
Stock No.	No. of Cores	Conductor			Thickness of Insulation System(Nom)	Diameter Over protective layer and Lacquer(Nom)	No. of Cores per layer	Sheath thickness(Nom)	Diameter over sheath (Nom)	Screen Wire Diameter(Nom)	Overall Diameter(Nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (Typical) x Single Wire Max. Dia.	Diameter(Nom)									Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq mm	mm	mm	mm	mm		mm	mm	mm	mm	V	V	V	Ohm	Mega.ohm	m	V	A	kg / km
5013-6856A1S	7	10	80/0.41	4.1	0.9	6.3	F.7	1.8	25.0	0.3	27.3	2000	440	500	1.95	100	100	440	60.0	1767
5013-6856S	7	16	126/0.41	5.4	0.9	7.6	F.7	1.8	29.4	0.3	31.7	2000	440	500	1.24	100	100	440	84.0	2185
5013-6857S	7	25	189/0.41	6.8	1.1	9.0	F.7	2.0	34.5	0.4	37.5	3500	440	500	0.795	55	100	440	99.0	3262
5013-6858S	7	35	259/0.41	7.8	1.1	10.0	F.7	2.1	38.0	0.4	41.1	3500	440	500	0.565	45	100	440	120	3963
5013-6859S	7	50	378/0.41	9.5	1.1	11.7	F.7	2.3	44.1	0.4	47.3	3500	440	500	0.393	45	100	440	150	4512
5013-6860S	7	70	348/0.51	11.5	1.1	13.7	F.7	2.5	51.2	0.4	54.5	3500	440	500	0.277	40	100	440	189	6506
5013-6861A1S	14	2.5	50 / 0.26	2	0.6	3.6	F.4.10	1.5	19.3	0.3	22.0	2000	440	500	8.21	100	100	440	21.5	976
5013-6861S	14	4	56 / 0.31	2.5	0.6	4.1	F.4.10	1.5	21.5	0.3	24.0	2000	440	500	5.09	100	100	440	27.5	1183
5013-6862S	14	6	84/0.31	3.1	0.7	4.9	F.4.10	1.7	25.5	0.3	27.8	2000	440	500	3.39	100	100	440	39.5	1562
5013-6863S	14	10	80/0.41	4.1	0.9	6.3	F.4.10	1.9	32.0	0.4	35.7	2000	440	500	1.95	100	100	440	50.0	2800
5013-6864S	14	16	126/0.41	5.4	0.9	7.6	F.4.10	2.1	38.2	0.4	41.2	2000	440	500	1.24	100	100	440	70.0	3850
5013-6865S	14	25	189/0.41	6.8	1.1	9.0	F.4.10	2.4	45.0	0.4	48.1	3500	440	500	0.795	55	100	440	82.5	5519
5013-6866S	14	35	259/0.41	7.8	1.1	10.0	F.4.10	2.5	49.6	0.4	53.2	3500	440	500	0.565	45	100	440	100.0	6469
5013-6867A1S	18	4	56 / 0.31	2.5	0.6	4.1	F.6.12	1.8	24.5	0.3	26.4	2000	440	500	5.09	100	100	440	24.8	1722
5013-6867S	18	6	84 / 0.31	3.1	0.7	4.9	F.6.12	1.8	28.5	0.3	30.9	2000	440	500	3.39	100	100	440	35.6	2077
5013-6868S	18	10	80 / 0.41	4.1	0.7	5.9	F.6.12	2.1	34.1	0.4	39.8	2000	440	500	1.95	100	100	440	45.0	4315
5013-6869S	18	16	126/0.41	5.4	0.8	7.6	F.6.12	2.3	43.0	0.4	46.2	2000	440	500	1.24	100	100	440	63.0	4896
5013-6870A1S	30	1	32/0.21	1.32	0.5	2.7	F.4.10.16	1.6	21.1	0.3	23.5	2000	440	500	20.0	100	100	440	9.2	1032
5013-6870S	30	1.5	30 / 0.26	1.55	0.5	3.0	F.4.10.16	1.6	22.5	0.3	25.1	2000	440	500	13.7	100	100	440	13.2	1185
5013-6871S	30	2.5	50 / 0.26	2	0.6	3.6	F.4.10.16	1.7	26.9	0.3	29.2	2000	440	500	8.21	100	100	440	17.2	1643
5013-6872S	30	4	56 / 0.31	2.5	0.6	4.1	F.4.10.16	1.9	30.5	0.4	34.1	2000	440	500	5.09	100	100	440	22.0	2368
5013-6873S	30	6	84 / 0.31	3.1	0.7	4.9	F.4.10.16	2.1	36.1	0.4	39.1	2000	440	500	3.39	100	100	440	31.6	3194
5013-6874S	30	10	80 / 0.41	4.1	0.7	5.9	F.4.10.16	2.5	43.3	0.4	50.0	2000	440	500	1.95	100	100	440	40.0	5447

Note: If the Copper wire braiding is required below the sheath, then for sizes 35 Sq. mm and above the diameter will be increased by 1 mm

TABLES OF DIMENSIONS**PART C****TABLE C13: MULTI CORE CABLES, OVERALL SCREENED**

1	2	3			4	5	6	7	8	9	10	11	12		13	14	15	16		17
Stock No.	No. of Cores	Conductor			Thickness of Insulation System(Nom)	Diameter Over protective layer and Lacquer(Nom)	No. of Cores per layer	Sheath thickness(Nom)	Diameter over sheath (Nom)	Screen Wire Diameter(Nom)	Overall Diameter(Nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (Typical) x Single Wire Max. Dia.	Diameter(Nom)									Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq mm	mm	mm	mm	mm			mm	mm	mm	V	V	V	Ohm	Mega.ohm	m	V	A	kg / km
5013-6828A1S	2	0.75	24 / 0.21	1.1	0.5	2.5	2	1.2	7.8	0.3	9.7	2000	440	500	26.7	100	250	440	13.6	145
5013-6840A1S	3	0.75	24 / 0.21	1.1	0.5	2.5	3	1.2	8.2	0.3	10.2	2000	440	500	26.7	100	250	440	11.9	163
5013-6877A1S	4	0.75	24 / 0.21	1.1	0.5	2.5	F.4	1.2	8.9	0.3	10.8	2000	440	500	26.7	100	250	440	11.9	204
5013-6878A1S	5	0.75	24 / 0.21	1.1	0.5	2.5	F.5	1.2	9.6	0.3	11.8	2000	440	500	26.7	100	250	440	10.3	235
5013-6879A1S	7	0.75	24 / 0.21	1.1	0.5	2.5	F.7	1.2	11.3	0.3	13.2	2000	440	500	26.7	100	250	440	10.3	286
5013-6880A1S	10	0.75	24 / 0.21	1.1	0.5	2.5	2.8	1.2	12.8	0.3	14.9	2000	440	500	26.7	100	250	440	8.5	339
5013-6881A1S	14	0.75	24 / 0.21	1.1	0.5	2.5	F.4.10	1.3	14.1	0.3	16.1	2000	440	500	26.7	100	250	440	8.5	450
5013-6882A1S	18	0.75	24 / 0.21	1.1	0.5	2.5	F.6.12	1.3	15.5	0.3	17.7	2000	440	500	26.7	100	250	440	7.6	518
5013-6883A1S	24	0.75	24 / 0.21	1.1	0.5	2.5	2.8.14	1.4	18.2	0.3	20.4	2000	440	500	26.7	100	250	440	7.6	696
5013-6884A1S	30	0.75	24 / 0.21	1.1	0.5	2.5	F.4.10.16	1.5	19.5	0.3	21.9	2000	440	500	26.7	100	250	440	6.8	822
5013-6885A1S	36	0.75	24 / 0.21	1.1	0.5	2.5	F.6.12.18	1.5	20.9	0.3	23.3	2000	440	500	26.7	100	250	440	6.8	960
5013-6886A1S	44	0.75	24 / 0.21	1.1	0.5	2.5	2.8.14.20	1.6	23.6	0.3	26.1	2000	440	500	26.7	100	250	440	6.8	1139
5013-6828S	2	1	32 / 0.21	1.3	0.5	2.7	2	1.2	8.2	0.3	10.1	2000	440	500	20	100	250	440	18.4	154
5013-6840S	3	1	32 / 0.21	1.3	0.5	2.7	3	1.2	8.6	0.3	10.6	2000	440	500	20	100	250	440	16.1	176
5013-6871S	4	1	32 / 0.21	1.3	0.5	2.7	F.4	1.2	9.3	0.3	11.3	2000	440	500	20	100	250	440	16.1	217
5013-6878S	5	1	32 / 0.21	1.3	0.5	2.7	F.5	1.2	10.1	0.3	12.3	2000	440	500	20	100	250	440	13.8	254
5013-6879S	7	1	32 / 0.21	1.3	0.5	2.7	F.7	1.2	12.0	0.3	13.9	2000	440	500	20	100	250	440	13.8	308
5013-6880S	10	1	32 / 0.21	1.3	0.5	2.7	2.8	1.2	13.6	0.3	15.7	2000	440	500	20	100	250	440	11.5	365

TABLES OF DIMENSIONS

PART C

TABLE C13: MULTI CORE CABLES, OVERALL SCREENED

1	2	3			4	5	6	7	8	9	10	11	12		13	14	15	16		17
Stock No.	No. of Cores	Conductor			Thickness of Insulation System(Nom)	Diameter Over protective layer and Lacquer(Nom)	No. of Cores per layer	Sheath thickness(Nom)	Diameter over sheath (Nom)	Screen Wire Diameter(Nom)	Overall Diameter(Nom)	Test voltage	Gas Flame Test		Maximum conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (Typical) x Single Wire Max. Dia.	Diameter(Nom)									Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq mm	mm	mm	mm	mm			mm	mm	mm	V	V	V	Oh m	Mega.oh m	m	V	A	kg / km
5013-6881S	14	1	32 / 0.21	1.3	0.5	2.7	F.4.10	1.3	14.9	0.3	17.0	2000	440	500	20	100	250	440	11.5	486
5013-6882S	18	1	32 / 0.21	1.3	0.5	2.7	F.6.12	1.3	16.5	0.3	18.7	2000	440	500	20	100	250	440	10.3	551
5013-6883S	24	1	32 / 0.21	1.3	0.5	2.7	2.8.14	1.4	19.4	0.3	21.6	2000	440	500	20	100	250	440	10.3	733
5013-6884S	30	1	32 / 0.21	1.3	0.5	2.7	F.4.10.16	1.5	20.7	0.3	23.2	2000	440	500	20	100	250	440	9.2	859
5013-6885S	36	1	32 / 0.21	1.3	0.5	2.7	F.6.12.18	1.5	22.3	0.3	24.7	2000	440	500	20	100	250	440	9.2	1001
5013-6886S	44	1	32 / 0.21	1.3	0.5	2.7	2.8.14.20	1.6	25.2	0.3	27.7	2000	440	500	20	100	250	440	9.2	1188
5013-6829S	2	1.5	30 / 0.26	1.6	0.5	2.95	2	1.2	8.7	0.3	10.6	2000	440	500	13.7	100	250	440	26.4	223
5013-6841S	3	1.5	30 / 0.26	1.6	0.5	2.95	3	1.2	9.2	0.3	11.1	2000	440	500	13.7	100	250	440	23.1	240
5013-6889A1S	4	1.5	30 / 0.26	1.6	0.5	2.95	F.4	1.2	9.9	0.3	11.9	2000	440	500	13.7	100	250	440	23.1	280
5013-6890A1S	5	1.5	30 / 0.26	1.6	0.5	2.95	F.5	1.2	10.8	0.3	13.0	2000	440	500	13.7	100	250	440	19.8	310
5013-6891A1S	7	1.5	30 / 0.26	1.6	0.5	2.95	F.7	1.3	13.0	0.3	15.0	2000	440	500	13.7	100	250	440	19.8	410
5013-6892A1S	10	1.5	30 / 0.26	1.6	0.5	2.95	2.8	1.4	15.0	0.3	17.1	2000	440	500	13.7	100	250	440	16.5	525
5013-6893A1S	14	1.5	30 / 0.26	1.6	0.5	2.95	F.4.10	1.4	16.2	0.3	18.3	2000	440	500	13.7	100	250	440	16.5	620
5013-6894A1S	18	1.5	30 / 0.26	1.6	0.5	2.95	F.6.12	1.5	18.2	0.3	20.4	2000	440	500	13.7	100	250	440	14.8	800
5013-6895A1S	24	1.5	30 / 0.26	1.6	0.5	2.95	2.8.14	1.7	21.5	0.3	23.7	2000	440	500	13.7	100	250	440	14.8	1010
5013-6896A1S	30	1.5	30 / 0.26	1.6	0.5	2.95	F.4.10.16	1.7	22.7	0.3	25.2	2000	440	500	13.7	100	250	440	13.2	1150
5013-6897A1S	36	1.5	30 / 0.26	1.6	0.5	2.95	F.6.12.18	1.8	24.7	0.3	27.1	2000	440	500	13.7	100	250	440	13.2	1350
5013-6898A1S	44	1.5	30 / 0.26	1.6	0.5	2.95	2.8.14.20	2	28.0	0.3	30.5	2000	440	500	13.7	100	250	440	13.2	1700

TABLES OF DIMENSIONS**PART C****TABLE C13: MULTI CORE CABLES, OVERALL SCREENED**

1	2	3			4	5	6	7	8	9	10	11	12		13	14	15	16		17
Stock No.	No. of Cores	Conductor			Thickness of Insulation System(Nom)	Diameter Over protective layer and Lacquer(Nom)	No. of Cores per layer	Sheath thickness(Nom)	Diameter over sheath (Nom)	Screen Wire Diameter(Nom)	Overall Diameter(Nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (Typical) x Single Wire Max. Dia.	Diameter(Nom)									Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq mm	No x mm	mm	mm	mm		mm	mm	mm	mm	V	V	V	Ohm	Mega.ohm	m	V	A	kg / km
5013-6830S	2	2.5	50 / 0.26	2	0.6	3.6	2	1.2	10.0	0.3	12.0	2000	440	500	8.21	100	250	440	34.4	262
5013-6842S	3	2.5	50 / 0.26	2	0.6	3.6	3	1.2	10.6	0.3	12.6	2000	440	500	8.21	100	250	440	30.1	304
5013-6889S	4	2.5	50 / 0.26	2	0.6	3.6	F.4	1.2	11.5	0.3	13.5	2000	440	500	8.21	100	250	440	30.1	362
5013-6890S	5	2.5	50 / 0.26	2	0.6	3.6	F.5	1.2	12.5	0.3	14.5	2000	440	500	8.21	100	250	440	25.8	439
5013-6891S	7	2.5	50 / 0.26	2	0.6	3.6	F.7	1.3	15.2	0.3	17.2	2000	440	500	8.21	100	250	440	25.8	559
5013-6892S	10	2.5	50 / 0.26	2	0.6	3.6	2.8	1.4	17.6	0.3	19.7	2000	440	500	8.21	100	250	440	21.5	705
5013-6893S	14	2.5	50 / 0.26	2	0.6	3.6	F.4.10	1.4	19.1	0.3	21.2	2000	440	500	8.21	100	250	440	21.5	871
5013-6894S	18	2.5	50 / 0.26	2	0.6	3.6	F.6.12	1.5	21.4	0.3	23.6	2000	440	500	8.21	100	250	440	19.3	1105
5013-6895S	24	2.5	50 / 0.26	2	0.6	3.6	2.8.14	1.7	25.4	0.3	27.6	2000	440	500	8.21	100	250	440	19.3	1401
5013-6871S	30	2.5	50 / 0.26	2	0.6	3.6	F.4.10.16	1.7	26.9	0.3	29.1	2000	440	500	8.21	100	250	440	17.2	1643
5013-6897S	36	2.5	50 / 0.26	2	0.6	3.6	F.6.12.18	1.8	29.2	0.4	32.2	2000	440	500	8.21	100	250	440	17.2	1954
5013-6898S	44	2.5	50 / 0.26	2	0.6	3.6	2.8.14.20	2	33.2	0.4	36.2	2000	440	500	8.21	100	250	440	17.2	2443

TABLES OF DIMENSIONS

PART C

TABLE C13: MULTI CORE CABLES, OVERALL SCREENED

1	2	3			4	5	6	7	8	9	10	11	12		13	14	15	16		17
Stock No.	No. of Cores	Conductor			Thickness of Insulation System(Nom)	Diameter Over protective layer and Lacquer(Nom)	No. of Cores per layer	Sheath thickness(Nom)	Diameter over sheath (Nom)	Screen Wire Diameter(Nom)	Overall Diameter(Nom)	Test voltage	Gas Flame Test		Maximum Conductor Resistance per km at 20°C	Minimum Insulation Resistance per km at 20°C at 500 V	Length on drum	Rating		Approx. Weight of the cable
		Nominal Area	No. of Wires (Typical) x Single Wire Max. Dia.	Diameter(Nom)									Test Voltage	Minimum Breakdown Voltage				Voltage	Current	
		Sq mm	No x mm	mm	mm	mm		mm	mm	mm	mm	V	V	V	Ohm	Mega.ohm	m	V	A	kg / km
5013-6831S	2	4	56 / 0.31	2.6	0.6	4.15	2	1.2	11.1	0.3	13.1	2000	440	500	5.09	100	250	440	44.0	321
5013-6843S	3	4	56 / 0.31	2.6	0.6	4.15	3	1.2	11.8	0.3	13.8	2000	440	500	5.09	100	250	440	38.5	370
5013-6901S	4	4	56 / 0.31	2.6	0.6	4.15	F.4	1.2	12.8	0.3	14.8	2000	440	500	5.09	100	250	440	38.5	449
5013-6902S	5	4	56 / 0.31	2.6	0.6	4.15	F.5	1.2	14.0	0.3	16.0	2000	440	500	5.09	100	250	440	33.0	552
5013-6903S	7	4	56 / 0.31	2.6	0.6	4.15	F.7	1.4	17.3	0.3	19.4	2000	440	500	5.09	100	250	440	33.0	739
5013-6904S	10	4	56 / 0.31	2.6	0.6	4.15	2.8	1.5	20.0	0.3	22.2	2000	440	500	5.09	100	250	440	27.5	934
5013-6861S	14	4	56 / 0.31	2.6	0.6	4.15	F.4.10	1.5	21.7	0.3	23.9	2000	440	500	5.09	100	250	440	27.5	1185
5013-6906S	18	4	56 / 0.31	2.6	0.6	4.15	F.6.12	1.6	24.4	0.3	26.6	2000	440	500	5.09	100	250	440	24.7	1472
5013-6907S	24	4	56 / 0.31	2.6	0.6	4.15	2.8.14	1.8	28.9	0.3	31.1	2000	440	500	5.09	100	250	440	24.7	1871
5013-6872S	30	4	56 / 0.31	2.6	0.6	4.15	F.4.10.16	1.9	30.8	0.4	33.8	2000	440	500	5.09	100	250	440	22.0	2372
5013-6909S	36	4	56 / 0.31	2.6	0.6	4.15	F.6.12.18	2	33.5	0.4	36.5	2000	440	500	5.09	100	250	440	22.0	2764